Fisher-Glass Learning: Classification Bounds and Subpopulation Audits

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Abstract

High predictive accuracy can coexist with sensitivity to task-preserving nuisance changes. Building on the Fisher-glass framework, we derive local classification lower bounds and develop subpopulation audits based on nuisance-projected information. The bounds connect efficient task information to minimax excess classification risk under explicit distinguishability and decision-separation conditions. We derive regularization-aware audit bounds and evaluate frozen population gates on held-out data. A prospective Waterbirds study uses an uncentered feature–Schur diagnostic constructed from task logits and background-probe scores to select an eligible validation group. Across five training seeds, interventions preserving the extracted bird foreground and labels yield 93.2% accuracy under same-type controls and 40.6% under opposite-type swaps, a 52.6-percentage-point difference. Search using task labels and a fixed nuisance direction also identifies held-out populations with high accuracy and low estimated projected predictive Fisher information. Training and acquisition experiments evaluate changes in information and predictive performance together. The resulting framework pairs predictive performance with explicit measures of task information and nuisance alignment.

1 Introduction

Reliable prediction under distribution shift requires understanding the evidence supporting a model's decisions. A classifier can perform well on a subpopulation by using a nuisance cue correlated with its label, then fail when that correlation changes. Group-robust learning and error-based reweighting make subpopulation performance central to robust machine learning [Sagawa et al., 2020, Liu et al., 2021]. We ask a complementary question: *how much task information remains after nuisance variation is accounted for?*

We investigate this question through a prospective Waterbirds intervention. Using validation data, we select an eligible group by the uncentered feature–Schur fraction of task-logit and background-probe coordinates. Models and the selected group are frozen before held-out evaluation. Across five training seeds, the selected group achieves 92.0% accuracy on untouched images. Background replacements preserving the extracted bird foreground and labels yield 93.2% accuracy under same-type controls and 40.6% under opposite-type swaps, a controlled difference of 52.6 percentage points. This comparison quantifies sensitivity to the audited background change within an accurately classified population.

Efficient-score theory describes the task information remaining after nuisance adjustment through an orthogonal projection [Bickel et al., 1993, Cox and Reid, 1987]. The Fisher-glass framework of Mansouri and Arai [2026] connects this local geometry to inverse-information tails and a population tail certificate. We build on this framework to study classification difficulty and the evaluation of

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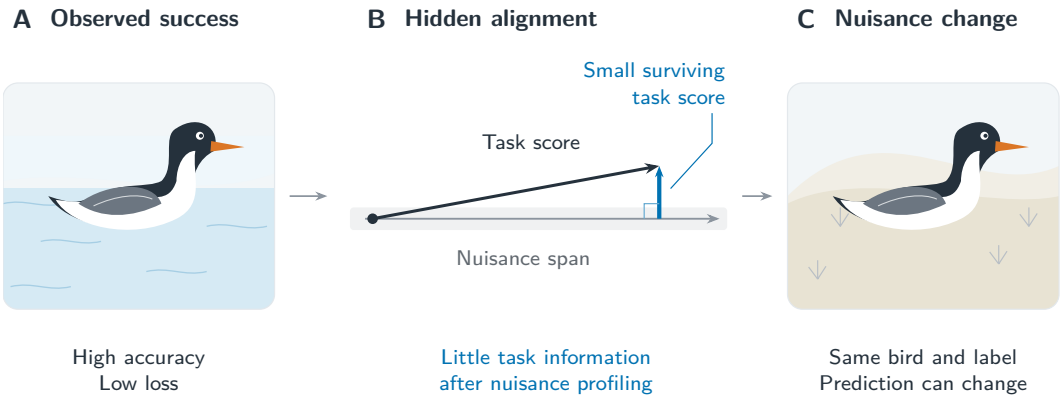


Figure 1: **Conceptual motivation for auditing task information under nuisance variation.** A nuisance cue can support high accuracy and low loss (A), while task–nuisance alignment leaves little information after projection (B). A task-preserving background change can expose the resulting sensitivity (C). The identical vector bird is reused in both environments; these are conceptual illustrations, not experimental images.

trained predictors. Figure 1 introduces the geometric motivation, and Section 2 specifies the task and nuisance coordinates.

Our classification result combines local indistinguishability with quantitative Bayes-decision separation. The resulting lower bound retains the environment-dependent information, separation constants, and local radius. We then develop the statistical ingredients needed for an operational audit: regularization-aware information bounds, supported population gates, and independent evaluation of selected subpopulations. Given task labels and a fixed nuisance direction, the search identifies a held-out Waterbirds population with 98.8% accuracy and low estimated projected predictive Fisher information.

Our contributions are:

- **Local classification theory.** We derive an excess-classification-risk lower bound under explicit distinguishability and decision-separation conditions, connecting efficient task information to local learning difficulty.
- **Statistical audit methodology.** We derive bounds for regularized Schur estimates and independently evaluated population gates, accounting for membership normalization and support correction.
- **Machine-learning evaluation.** We study controlled background sensitivity, held-out discovery of accurate low-information populations, and the joint information–performance outcomes of targeted training and acquisition.

2 Task–Nuisance Geometry and Information Collapse

Let $e \sim \Pi$ index an observed group or a latent subpopulation. We fix the task contrast, admissible nuisance variations, and a common normalization of task coordinates throughout the audit. Information then describes the task distinction remaining after adjustment for those nuisance variations.

2.1 Task Information After Nuisance Profiling

Consider a local model $P_{\theta,\mu,e}$ with task parameter θ and nuisance parameter μ. A likelihood score measures the local change in log likelihood as a parameter varies. At $P_e = P_{0,0,e}$, assume differentiability in quadratic mean, with centered, square-integrable task-score vector $s_{T,e}$ and nuisance-score vector $s_{N,e}$. Let $\mathcal{T}_{N,e}$ be the closed span of admissible nuisance scores in $L^2(P_e)$, with inner product $\langle f, g\rangle_e = \mathbb{E}_{P_e}[fg]$, and let $\Pi_{\mathcal{T}_{N,e}}$ denote orthogonal projection onto that span.

The **efficient task score** and its information are

$$\begin{aligned} s^{\mathrm{eff}}_{T,e} &= s_{T,e} - \Pi_{\mathcal{T}_{N,e}} s_{T,e}, \\ F_e := F^{\mathrm{eff}}_{T|N,e} &= \mathbb{E}_{P_e}[s^{\mathrm{eff}}_{T,e}(s^{\mathrm{eff}}_{T,e})^\top] = J_{TT,e} - J_{TN,e} J^{-1}_{NN,e} J_{NT,e}. \end{aligned} \tag{1}$$

The projection acts componentwise. The Schur expression applies when finite nuisance scores span $\mathcal{T}_{N,e}$ and $J_{NN,e} \succ 0$, with $J_{AB,e} = \mathbb{E}_{P_e}[s_{A,e} s_{B,e}^\top]$ [Bickel et al., 1993, Tsiatis, 2006, Cox and Reid, 1987]. Write $\lambda_e = \lambda_{\min}(F_e)$ for the weakest task information. A zero eigenvalue identifies a task direction contained in the nuisance span.

For scalar task and nuisance scores with positive raw information, their Fisher correlation $\rho_e = J_{TN,e}/\sqrt{J_{TT,e} J_{NN,e}}$ gives

$$F_e = J_{TT,e}(1 - \rho_e^2), \qquad \underbrace{F_e / J_{TT,e}}_{\text{surviving-information fraction}} = 1 - \rho_e^2. \tag{2}$$

Absolute information measures remaining curvature in the declared coordinates. The fraction measures the proportion retained, lies in $[0, 1]$, and is invariant to nonzero scalar rescaling of either score. It compares alignment across slices with different raw information; its feature–Schur analogue supplies the prospective selection statistic. The model-level Fisher-Glass certificate, defined below, instead summarizes absolute information across environments.

2.2 Task–Nuisance Tangency and Loss–Information Separation

Let b_e, r_e be centered, orthogonal, unit-norm scores, with nuisance span $\mathrm{span}\{b_e\}$. For $\delta_e \geq 0$, the normalized construction gives

$$a_e = b_e + \delta_e r_e, \quad J_{TT,e} = 1 + \delta_e^2, \quad F_e = \delta_e^2, \quad \frac{F_e}{J_{TT,e}} = \frac{\delta_e^2}{1 + \delta_e^2}. \tag{3}$$

In the transverse-information geometry of Mansouri and Arai [2026], projection removes b_e, leaving $\delta_e r_e$. Raw information stays above one as efficient information vanishes quadratically; δ_e measures distance from the nuisance span. When the nuisance feature also supports small observed prediction risk, low loss coexists with arbitrarily weak efficient information. This separation concerns the evidence supporting prediction, even within accurately classified populations. The formal construction and proof appear in Appendix A.1. Figure 2 illustrates the geometry's population consequence: frequent near-tangencies produce a heavy inverse-information tail.

3 Tail Identifiability and Local Learning Difficulty

The frequency of near-null environments determines the inverse-information tail. Its classification consequence additionally depends on how weak task alternatives change the Bayes decision. We separate these ingredients before defining the certificate.

3.1 Collapse Geometry Determines Information Tails

In the tangency construction, $\lambda_e = \delta_e^2$: if $\Pr(\delta_e < t) \sim Ct^\gamma$, then $\Pr(\lambda_e^{-1} > x) \sim Cx^{-\gamma/2}$. The geometric tail law of Mansouri and Arai [2026] relates the inverse-information exponent to null-set codimension, information vanishing order, and nearby probability mass. We state the law with explicit regularity and localization assumptions.

Theorem 1 (Geometric tail law [Mansouri and Arai, 2026])**.** *Let $z \in \mathbb{R}^d$ parameterize environments with density p, and let the Fisher-null set $\mathcal{Z} = \{z : \lambda(z) = 0\}$ be a nonempty compact C^2 embedded submanifold without boundary, of codimension $c \in \{1, \ldots, d\}$. Suppose λ is continuous on the closure of a fixed tubular neighborhood of $\mathcal{Z}$ and bounded away from zero outside it. In normal coordinates $z = s + r\omega$, let $s \in \mathcal{Z}$, r be normal distance, and ω a unit normal direction. As $r \downarrow 0$, assume uniformly that information vanishes as $a(s,\omega) r^{2m}(1 + o(1))$ and density as $b(s,\omega) r^\nu (1 + o(1))$, where $m > 0$, $\nu > -c$, and the continuous coefficients are uniformly positive and bounded. Then, for some $C > 0$,*

$$\Pr(\lambda(z)^{-1} > x) \sim Cx^{-\beta}, \qquad \beta = \frac{c + \nu}{2m}. \tag{4}$$

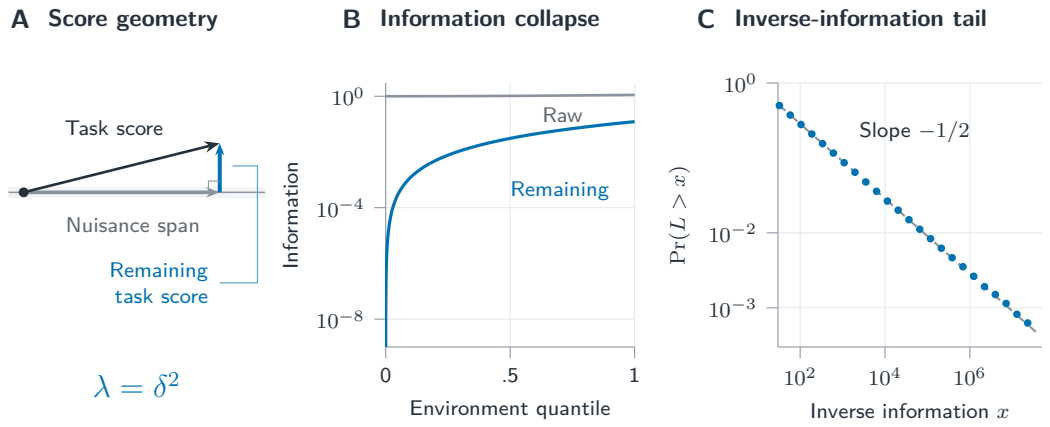


Figure 2: **Synthetic illustration of the geometric tail law** [Mansouri and Arai, 2026]. (A) The surviving score has norm δ and information $\lambda = \delta^2$. (B) Raw information stays above one while projected information approaches zero. (C) Empirical inverse-information survival (blue) follows the theoretical $x^{-1/2}$ tail (dashed). B–C share one regenerated seed-11 draw of 300,000 environments, $\delta \sim \mathrm{Uniform}(0, 0.35)$ and $L = (\delta^2 + 10^{-12})^{-1}$. The separately reported five-seed exponent is 0.498 ± 0.003 (nominal 95% normal-approximation interval, ± 1.96 standard errors).

An information threshold defines a narrow tube around $\mathcal{Z}$; integrating its probability mass determines the exponent. Greater codimension makes simultaneous collapse rarer, whereas higher-order vanishing produces a heavier tail. Quadratic collapse with positive density gives $\beta = c/2$, and the exact inverse-information moment is finite precisely when $\beta > 1$. Thus substantial information across most environments can coexist with divergent average inverse information. Appendix A.2 gives a complete proof under these assumptions, including the asymptotic constant.

3.2 Efficient Information and Local Classification Difficulty

The geometric tail acquires a classification consequence when nearby nuisance-adjusted alternatives induce quantitatively different optimal decisions. Define classification risk $R_P(g) = P\{g(X) \neq Y\}$, Bayes risk $R_P^\star = \inf_g R_P(g)$, and excess risk $\mathcal{E}_P(g) = R_P(g) - R_P^\star$. For a specified local family, let $\mathfrak{R}_e(n)$ denote minimax expected excess classification risk based on n independent observations from environment e.

Theorem 2 (Local classification difficulty)**.** *Fix $\lambda_e > 0$ and a unit weakest efficient-information direction u_e. For every $0 < t \leq t_{0,e}$, suppose the local family contains $P_{e,+t}, P_{e,-t}$ with task coordinates $\pm t u_e$ and nuisance coordinates allowed to vary. Assume, for constants $K_e, c_e, t_{0,e} > 0$ and $q > 0$,*

$$\mathrm{KL}(P_{e,+t} \| P_{e,-t}) \leq K_e t^2 \lambda_e, \qquad \inf_g \{\mathcal{E}_{P_{e,+t}}(g) + \mathcal{E}_{P_{e,-t}}(g)\} \geq c_e t^q. \tag{5}$$

Then, for a universal constant $c_0 > 0$,

$$\mathfrak{R}_e(n) \geq c_0 c_e \min\{t_{0,e}^q, (n K_e \lambda_e)^{-q/2}\}. \tag{6}$$

The KL condition limits distinguishability after nuisance adjustment; decision separation specifies the cost of confusing the alternatives. Both must hold over the environment-specific radius $t_{0,e}$. For quadratic separation, $q = 2$, the interior bound scales as $c_e/(n K_e \lambda_e)$. The minimum retains the finite scale of the local problem as information decreases. This is a conditional local result: efficient

information sets the testing scale, while boundary geometry supplies its classification meaning. The complete testing proof and its tail-aggregation implication are in Appendix A.3.

3.3 The Fisher-Glass Certificate

Upper-tail conditional value-at-risk, CVaR_α, averages the largest $1-\alpha$ fraction of a quantity, with fractional weighting at atoms [Rockafellar and Uryasev, 2000, 2002]. Following Mansouri and Arai [2026], we apply the Fisher-glass certificate to classification environments $e \sim \Pi$, using the weakest task information λ_e and stabilization ε:

$$L_e^{(\varepsilon)} = (\lambda_e + \varepsilon)^{-1}, \qquad I_\alpha^{\mathrm{FG}} = \left[\mathrm{CVaR}_{\alpha, e\sim\Pi}(L_e^{(\varepsilon)})\right]^{-1}, \quad 0 < \alpha < 1. \tag{7}$$

A larger certificate means a smaller inverse-information tail, expressed in information units. Stabilization $\varepsilon > 0$ caps inverse loss at $1/\varepsilon$; it does not remove the underlying null geometry. Under the theorem's power-law regime, the exact limit $\varepsilon \downarrow 0$ is positive for $\beta > 1$ and vanishes for $\beta \le 1$.

Proposition 1 (Information coverage)**.** *For $\rho > \varepsilon$,*

$$I_\alpha^{\mathrm{FG}} \ge \rho \quad \Longrightarrow \quad \Pr_{e\sim\Pi}(\lambda_e \ge \rho - \varepsilon) \ge \alpha. \tag{8}$$

The certificate therefore establishes an absolute information floor on at least an α-fraction of the specified environment distribution. Its connection to classification retains the local-radius and decision-separation factors in the preceding bound. The CVaR representation, coverage proof, and limiting interpretation appear in Appendix A.4. The operational audit estimates this information target on independently evaluated populations.

4 Computing and Auditing Fisher-Glass

An operational audit fixes the task contrast, nuisance hypothesis, matrix construction, and audit distribution before measuring information on independent data. We work in low-dimensional task and nuisance subspaces, so the reported information concerns these declared coordinates. Coordinate normalization is specified for each experiment; fitted nuisance probes and temperature calibration are frozen before final evaluation. The audit reports support and conditioning alongside its information values.

4.1 Operational Scores and Nuisance Coordinates

The audit uses a declared matrix target. Predictive Fisher is computed from likelihood scores under the fitted predictive distribution; observed-label score moments and unweighted feature moments define additional audit targets. The prospective Waterbirds experiment uses the uncentered feature–Schur fraction of fixed task-logit and background-probe coordinates. Hidden-slice discovery uses estimated temperature-calibrated predictive Fisher with class-specific shrinkage $\eta = 0.10$, retained during search and held-out evaluation. Unweighted feature geometry supplies a complementary comparison. Appendix B.1 specifies each construction.

The experiments prespecify nuisance coordinates from metadata, fitted annotation probes, or a fixed semantic direction. Without nuisance annotations, the specified semantic direction serves as the audited nuisance hypothesis. We report fitting splits, normalization, and nuisance-block conditioning; cross-validation assesses gate stability. Appendix D.5 records the checks performed. Formal anchor task-preservation tests and admission thresholds are criteria for extending this family, rather than implemented selection steps.

4.2 Stable Estimation and Exact-Information Audits

Let M_i be the per-example task–nuisance matrix under the chosen construction, with task and nuisance blocks indexed by T and N. For a predefined slice g of size n_g, form $\widehat{J}_g = n_g^{-1} \sum_{i\in g} M_i$ and compute the ridge Schur estimate

$$\widehat{F}_g^{\mathrm{rid}} = \widehat{J}_{TT,g} - \widehat{J}_{TN,g}(\widehat{J}_{NN,g} + \lambda_{\mathrm{rid}} I)^{-1} \widehat{J}_{NT,g}. \tag{9}$$

Algorithm 1 Fisher-Glass audit

1: **Specify:** predictor, matrix, coordinates, normalization, calibration, and audit weights.
2: **Discover and freeze:** predefined slices or constrained gates fitted on discovery data.
3: **Audit independently:** check support and conditioning; compute ridge, unregularized, and lower values.
4: **Report:** raw and projected information, surviving fraction, slice support, tail certificate, and uncertainty.

Real-data computations use $\lambda_{\mathrm{rid}} = 10^{-3}$. Ridge stabilizes residualization but raises information relative to the unregularized target. Let a_g bound estimation error and d_g bound the ridge-to-unregularized gap on a simultaneous event over the audited slices. For declared weights Π_G, use

$$\begin{aligned} \underline{\lambda}_g &= [\lambda_{\min}(\widehat{F}_g^{\mathrm{rid}}) - a_g - d_g]_+, \\ \underline{I}_{\alpha,G} &= \left[\mathrm{CVaR}_{\alpha,g\sim\Pi_G}\left((\underline{\lambda}_g + \varepsilon)^{-1}\right)\right]^{-1}. \end{aligned} \tag{10}$$

Here $[x]_+ = \max\{x, 0\}$. On that event, this lower-bounds the unregularized population certificate for the declared matrix and audit distribution. Certification also requires a positive lower confidence bound on nuisance conditioning. The implemented scalar audits use Bonferroni-adjusted Gaussian/Wald moment simulation at nominal 95% confidence, conditional on the fitted model and evaluated slices. Appendix B records the gap identity, error allowances, and numerical comparison.

4.3 Hidden-Slice Discovery and Held-Out Certification

A gate $h_\phi(x, y) \in [0, 1]$ assigns membership weights. Discovery maximizes inverse projected information subject to effective-sample-size, maximum-weight, and class-mass constraints, with shrinkage toward the same-class pooled matrix. Five validation folds assess transfer and stability; after fixing the protocol, a final gate is fitted on all discovery data, frozen, and evaluated on an untouched split. Its unshrunk target is the weighted ratio $J_\phi = \mathbb{E}[h_\phi M]/\mathbb{E}[h_\phi]$, with positive membership mass. Uncertainty must account for numerator, normalization, and any constraint-induced weight adjustment; the reported Waterbirds target retains the stated shrinkage. Appendix C specifies the constraints, objective, and held-out target.

5 Evaluating Information Audits

We test the diagnostic on a prospective nuisance intervention, isolate its geometry in synthetic experiments, and discover low-information populations on held-out data. The corresponding quantities are the surviving fraction, inverse-information tail, and predictive projected Fisher.

5.1 Prospective Diagnosis Under a Nuisance Intervention

On Waterbirds [Sagawa et al., 2020, Koh et al., 2021], the four groups cross bird class with background type and are written *class/background*. We fully fine-tune ImageNet-initialized MobileNetV3-small models [Howard et al., 2019] with empirical risk minimization. Validation data fit the temperature and cross-fitted background probe. Eligible slices have accuracy at least the overall validation accuracy and negative log-likelihood at most the median across slices. Among them, we select the smallest surviving fraction, $\lambda_{\min}(\widehat{F}_g^{\mathrm{rid}})/\widehat{J}_{TT,g}$. After pilot experiments, this rule is fixed for five additional training seeds, 22–26. Each checkpoint, temperature, probe, selected slice, and audit constant is frozen before intervention scoring.

The rule selects **waterbird/water in all five seeds**: its validation surviving fraction is 24.8%, compared with 59.0% for landbird/land, the other eligible slice. Held-out interventions recompose birds onto same-type or opposite-type backgrounds using the same extracted foreground layer, matte, and task labels. Same-type recomposition controls for the image-construction procedure. Selected-slice accuracy is 92.0% on original images, 93.2% under the control, and 40.6% under opposite-type replacement: a 52.6 ± 1.4 **percentage-point controlled drop**, with a nominal 95% normal-approximation interval across seeds.

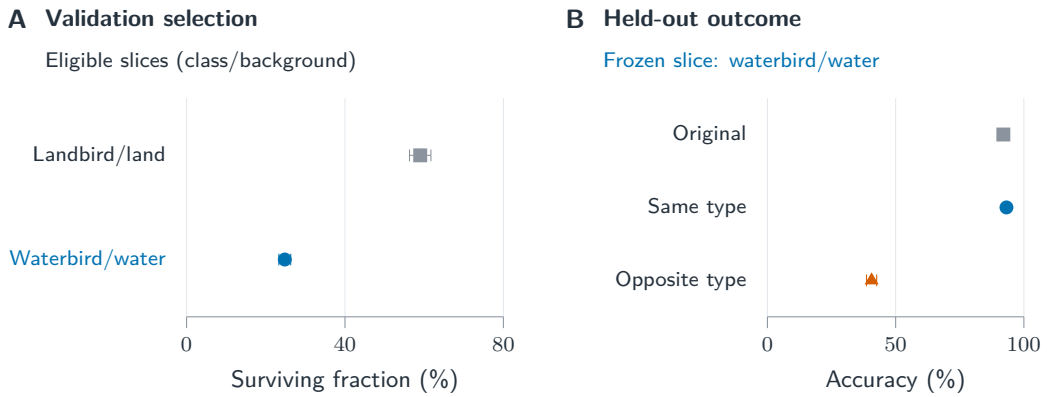


Figure 3: **Validation selection anticipates held-out background sensitivity.** (A) Waterbird/water has the lower uncentered feature–Schur fraction among eligible slices and is selected in all five seeds. Matched minimum-accuracy and maximum-NLL selectors choose the same group in all five. Eligibility and selection are evaluated separately within each seed; names denote class/background. (B) With this selection frozen, ERM accuracy is high on original images and same-type recompositions but falls under opposite-type background recomposition, a controlled difference of 52.6 percentage points. Intervals are nominal 95% normal approximations (± 1.96 standard errors) across seeds 22–26; the same-type arm controls for recomposition.

Within the eligible set, minimum surviving fraction, minimum validation accuracy, and maximum validation NLL select waterbird/water in all five seeds, with a 52.6-point controlled accuracy drop (Figure 3). Minimum raw and absolute projected information select landbird/land, with a 26.7-point drop. Centering switches selection in all five seeds, identifying slice means as a ranking contributor; predictive Fisher retains waterbird/water in all five. Appendix D.2 reports these checks, the complete selector comparison, and individual flip prediction.

5.2 Synthetic Classification and Tail Checks

Using nuisance feature s, orthogonal residual r, and task coordinate $z = s + \delta_e r$ as in Section 2, the evaluated classifier has logit $4s$. For near-null mass exponent $\gamma = 1$, five-seed mean accuracy is 94.54% and loss-$\mathrm{CVaR}_{0.95} = 0.262$. Yet $I^{\mathrm{FG}}_{0.95} = (1.22 \pm 0.53) \times 10^{-8}$ with $\varepsilon = 10^{-12}$, despite raw information bounded below by one.

The inverse-information exponent is 0.498 ± 0.003, matching $\beta = \gamma/2 = 0.5$ in the geometric tail law of Theorem 1 (Figure 2). Log-survival regression uses the largest 5% of inverse-information observations. Independent-branch and reserve experiments check the structural laws of Mansouri and Arai [2026], including $\beta = K/2$ for K branches. Uncertainties are nominal 95% normal-approximation intervals across five seeds. These measurements exhibit favorable prediction alongside weak projected information and verify the predicted inverse-information tail behavior. Appendix D.3 records the structural checks and additional summaries.

5.3 Discovering Low-Information Subpopulations

Discovery receives images, task labels, frozen ImageNet MobileNetV3-small representations, and a fixed CLIP water-versus-land nuisance direction [Radford et al., 2021]. Background and group annotations remain sealed through discovery and test evaluation. A class-conditional linear gate targets 10% within-class mass under the support constraints of Section 4.3. Five validation folds assess transfer; after fixing the protocol, the gate is refit on validation data, frozen, and evaluated once on the untouched test split.

The discovered slice has 98.8% **accuracy** and lower estimated projected predictive Fisher information than high-loss and random selectors with the same target within-class mass in every seed (Figure 4). Its temperature-calibrated predictive information is 0.005 ± 0.001 and loss 0.031 ± 0.017, with 95% Student-t intervals across five paired seeds. Accuracy and loss use support-corrected membership

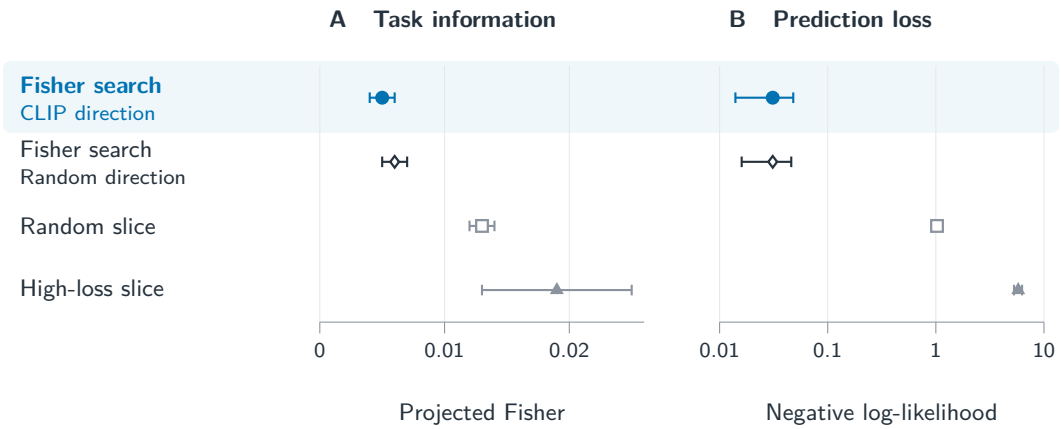


Figure 4: **Discovery finds accurate populations with low predictive information.** Held-out predictive projected Fisher (A) and negative log-likelihood (B; log scale), with means and 95% Student-t intervals across five paired seeds. The highlighted slice has 98.8% accuracy. Discovery uses task labels and a fixed nuisance direction; group annotations remain sealed and the gate is frozen before test evaluation. Predictive moments retain calibration and shrinkage 0.10. The random-direction control attains similar information.

weights without information pooling; Appendix C.1 defines these metrics and overlap conventions. Its top-10% Jaccard overlap with the high-loss slice is 0.001: information search finds a highly accurate population largely distinct from loss-based selection.

The random-direction control attains similar predictive information. The paired CLIP-minus-random interval and both unweighted comparisons with high-loss selection span zero. These unweighted comparisons reevaluate the predictive-selected gates under alternative matrices. The strongest observed separation therefore concerns predictive information, while the incremental contribution of nuisance semantics remains unresolved. Appendix D.4 gives the full table, paired intervals, support diagnostics, and revealed composition.

6 Information-Guided Interventions

The audit supplies objectives for data allocation, training, and acquisition. An observed-source allocation instantiates the Fisher-reserve principle of Mansouri and Arai [2026]: assigning fraction r to an anchor with unregularized projected information at least $\rho_0 I$ establishes the floor $r\rho_0$, under common coordinates and positive-definite nuisance blocks (Appendix E.1). Learning combines empirical task risk or Group DRO with inverse-information tail risk:

$$\min_{w} \ \widehat{R}_{\mathrm{base}}(w) + \lambda_{\mathrm{FG}} \, \mathrm{CVaR}_{\alpha,g}\left[\left\{\lambda_{\min}(\widehat{F}_g^{\mathrm{rid}}(w)) + \varepsilon\right\}^{-1}\right]. \tag{11}$$

Weak projected-direction penalties complement base task risk. Coordinates follow Section 4. Frozen-feature Waterbirds uses validation group labels to select the largest validation certificate among coefficients within two percentage points of the best validation worst-group accuracy.

Table 1: Frozen-feature Waterbirds training. Means with nominal 95% normal-approximation intervals (± 1.96 standard errors) across five seeds; ridge certificates use $\alpha = 0.95$. Validation selection follows Section 6.

Objective	Average accuracy (%)	Worst-group accuracy (%)	Fisher-Glass certificate
ERM	77.2 ± 0.1	55.7 ± 0.3	0.198 ± 0.001
Fisher-Glass	80.4 ± 0.2	56.3 ± 0.3	0.399 ± 0.006
Group DRO	80.9 ± 0.3	67.3 ± 0.3	0.233 ± 0.003
Group DRO + Fisher-Glass	80.6 ± 0.9	69.4 ± 1.3	0.614 ± 0.059

Fisher-Glass approximately doubles ERM's certificate (Table 1); with Group DRO it raises the certificate from 0.233 to 0.614 and mean worst-group accuracy from 67.3% to 69.4%. Stronger penalties can reduce accuracy (Appendix E.3), so selecting an operating point requires task and validation criteria.

Prospective training increases the validation certificate from 0.289 to 0.330, while the controlled background-swap drop is 52.9 points versus 52.6 for ERM. Constrained Fisher-only repair selects the unchanged model for the discovered slice in all five seeds. Post hoc selection increases hidden information but reduces worst-group accuracy by 12.6 points; coefficients coincide with validation-NLL minimization in every seed. Appendix E.3 records the comparisons.

For acquisition, candidate a has modified feature/input x_a^{cf}, assigned label y_a, and current predictive loss $\ell_w(x_a^{\mathrm{cf}}, y_a)$. *Tail-transverse influence* (TTI) multiplies local tail-information gain by this loss, $\mathrm{TTI}(a) = G_\alpha(a)\ell_w(x_a^{\mathrm{cf}}, y_a)$. Gain differentiates the Schur complement along its weakest task direction, weighting environments by inverse-information CVaR sensitivity (Appendix E.4). Waterbirds implements $\widetilde{\mathrm{TTI}}_\alpha(a) = \widetilde{G}_\alpha(a)\ell_w(x_a^{\mathrm{cf}}, y_a)$ with a normalized residual-score surrogate $\widetilde{G}_\alpha$; Appendix E.5 specifies its weighting and allocation conventions.

Waterbirds candidates translate features by a same-label background-group mean difference, scaled by γ_{cf}, using task and background annotations. Training data supply means and the nuisance probe; validation data supply candidates, selected features enter retraining, and evaluation is held out. Budgets count acquisitions relative to original training-set size. At unit feature-shift strength, $\gamma_{\mathrm{cf}} = 1$, and a 1% budget, $\widetilde{\mathrm{TTI}}_\alpha$ yields a mean worst-group accuracy gain of 4.52 points versus 3.27 for high-loss selection; high loss produces the larger mean certificate gain. At this strength, $\widetilde{\mathrm{TTI}}_\alpha$ has the larger mean worst-group accuracy gain at all three evaluated budgets; pure tail gain yields smaller improvements. See Appendix E.5 for construction and budgets.

7 Related Work

Efficient information and classification. Efficient-score theory defines task information after projection away from nuisance tangent spaces [Bickel et al., 1993, Newey, 1990, Tsiatis, 2006, Cox and Reid, 1987]. The Fisher-glass framework of Mansouri and Arai [2026] studies the environmental distribution of that information through geometric tail laws and inverse-information certification. We develop its classification consequences through local testing reductions [Le Cam, 1986, Tsybakov, 2009] and quantitative Bayes-decision separation [Audibert and Tsybakov, 2007]. The operational analysis addresses Schur estimation, ridge effects, and independent evaluation of learned population gates.

Tail risk, robust learning, and gradient geometry. CVaR concentrates objectives on adverse outcomes [Rockafellar and Uryasev, 2000, 2002]; Group DRO applies robust optimization to predictive loss across groups [Sagawa et al., 2020]. Invariant Risk Minimization seeks shared optimal classifiers [Arjovsky et al., 2019], while Fishr regularizes domain-level gradient variances [Rame et al., 2022]. We use inverse projected-information tail risk as a training penalty alongside empirical risk or Group DRO, and evaluate information and predictive performance together.

Hidden stratification and slice discovery. Hidden-stratification research motivates finer population evaluation [Oakden-Rayner et al., 2020]. GEORGE clusters representations to infer subclasses [Sohoni et al., 2020], JTT upweights initially misclassified examples [Liu et al., 2021], and AGRO learns adversarial group assignments using prediction loss [Paranjape et al., 2023]. Our search targets low predictive projected information given task labels and a fixed nuisance direction. Support constraints and an independently evaluated frozen gate make accurately classified, low-loss populations eligible audit targets.

Information-based design and acquisition. E-optimal design emphasizes the smallest information eigenvalue [Pukelsheim, 2006]; BAIT uses Fisher information for neural active learning [Ash et al., 2021]. TTI measures local information gain along a weak nuisance-adjusted task direction, weights environments by tail sensitivity, and multiplies by current task loss. Acquisition experiments assess realized information and accuracy changes after retraining.

8 Limitations

Specification and attribution. The audit depends on task contrast, nuisance space, and representation; nuisance variation must preserve the task distinction. Absolute eigenvalues depend on task scaling; ridge estimates also on nuisance scaling. Standardization supplies a comparison convention, not representation invariance. Predictive Fisher weights confidence; unweighted Schur statistics describe residual feature geometry. Their relation to population efficient information depends on the score model. Centering reverses the prospective ranking in all five seeds, so its original uncentered diagnostic depends partly on slice means. Comparable CLIP and random-direction discovery results and unweighted-control intervals containing zero leave semantic nuisance projection's incremental contribution unresolved. Confidence-matched comparisons and systematic nuisance controls remain necessary. Audits may expose subgroup failures, but treating a local certificate as a deployment guarantee could overlook harms outside the audited populations.

Local theory and audit scope. Efficient information, separation strength, and local radius jointly determine classification bounds, which require regularity and quantitative Bayes-decision separation. Sample size counts the audited environment's observations. Latent-environment pooling also depends on prevalence, shared information, and discovery. Coverage is relative to the declared environment distribution. Finite audits cover included groups or frozen slice rules; small mass and weak nuisance conditioning amplify uncertainty. Implemented Bonferroni-adjusted Gaussian/Wald audits have nominal coverage conditional on fitted model and evaluated family. The reported Wald procedure holds realized normalizations fixed and does not propagate uncertainty from normalizers estimated on evaluation data. Repeated searches must account for certification-data reuse. Empirical sample-size scaling and achievable classification rates remain open.

Development history and empirical breadth. After pilot experiments, the prospective rule was frozen for fresh training seeds on the same Waterbirds benchmark. Repetitions measure variability within this development setting; independent confirmatory datasets are needed. The prospective experiment establishes controlled background sensitivity in the selected group; comparative selection gains require additional matched evaluations. The probe's background attribute defines the intervention; same-type recomposition controls shared construction effects. Evidence centers on MobileNetV3-small image classifiers, TF-IDF text diagnostics, and synthetic constructions. Broader representations and entangled task–nuisance settings need evaluation. Unregularized audits used fresh training runs; checkpoint-specific conclusions concern those replications.

Information, performance, and intervention scope. Information gains do not establish behavioral repair: prospective swap sensitivity remains similar, constrained hidden-slice repair selects unchanged models, and post hoc selection increases hidden information while reducing worst-group accuracy. Prospective models lack accuracy matching in two of five seeds; post hoc choices coincide with validation-NLL minimization. Deployment requires prospective validation against its performance objective. Reserves assume an informative anchor and stated information-accumulation design. TTI measures the current predictor's local sensitivity; finite interventions and retraining are empirical tests. Transfer of annotation-conditioned acquisition feature shifts to realizable label-preserving interventions and other candidate families remains unestablished.

9 Conclusion

We develop classification and auditing methods for evaluating nuisance-projected task information in trained predictors. A local excess-risk lower bound connects efficient information to learning difficulty under explicit decision-separation conditions. Regularization-aware bounds and independent evaluation support audits of predefined groups and learned subpopulations. On Waterbirds, a validation-selected group exhibits a 52.6-point controlled accuracy drop under background replacement, while held-out discovery identifies accurate populations with low estimated projected predictive Fisher information. Training and acquisition experiments characterize the joint response of information and predictive performance. These results support evaluating classifiers through observed errors, specified information targets, and sensitivity to declared nuisance changes.

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A Proofs of the Core Results

This appendix establishes the results in Sections 2–3. We first derive the nuisance-projected information and its tangency construction. We then prove the inverse-information tail law, the local classification lower bound, and the information-coverage property of Fisher-Glass. Throughout, task coordinates use the normalization fixed in the main text, and environmental aggregation is with respect to the declared distribution Π.

A.1 Projection, Normalization, and Task–Nuisance Tangency

Fix an environment e and its reference distribution P_e. Let

$$s_{T,e} \in L_0^2(P_e)^{d_T}, \qquad s_{N,e} \in L_0^2(P_e)^{d_N}$$

be centered, square-integrable task and nuisance scores. Write

$$J_{AB,e} = \mathbb{E}_{P_e}\left[s_{A,e} s_{B,e}^\top\right], \qquad A, B \in \{T, N\}.$$

Assume that the coordinates of $s_{N,e}$ span the specified nuisance tangent space $\mathcal{T}_{N,e}$ and that $J_{NN,e}$ is positive definite.

Efficient-score projection. Set $\Gamma_e = J_{TN,e} J_{NN,e}^{-1}$. The residual satisfies

$$\mathbb{E}_{P_e}\left[(s_{T,e} - \Gamma_e s_{N,e}) s_{N,e}^\top\right] = J_{TN,e} - \Gamma_e J_{NN,e} = 0. \tag{A.1}$$

Because $\Gamma_e s_{N,e}$ lies componentwise in $\mathcal{T}_{N,e}$, this orthogonality identifies it as the componentwise $L^2(P_e)$ projection of the task score. Consequently,

$$\begin{aligned} s_{T,e}^{\mathrm{eff}} &= s_{T,e} - J_{TN,e} J_{NN,e}^{-1} s_{N,e}, \\ F_e &= \mathbb{E}_{P_e}\left[s_{T,e}^{\mathrm{eff}} (s_{T,e}^{\mathrm{eff}})^\top\right] \\ &= J_{TT,e} - J_{TN,e} J_{NN,e}^{-1} J_{NT,e}. \end{aligned} \tag{A.2}$$

This is the efficient-information construction used in Equation (1) [Bickel et al., 1993, Tsiatis, 2006]. Equivalently, for every task direction u,

$$\begin{aligned} u^\top F_e u &= \min_{v \in \mathbb{R}^{d_N}} \mathbb{E}_{P_e}\left[(u^\top s_{T,e} - v^\top s_{N,e})^2\right], \\ v^\star &= J_{NN,e}^{-1} J_{NT,e} u. \end{aligned} \tag{A.3}$$

Thus $F_e \succeq 0$. A nonzero direction u has zero efficient information exactly when $u^\top s_{T,e}$ belongs to the nuisance span in $L^2(P_e)$. The quantity $\lambda_e = \lambda_{\min}(F_e)$ measures the weakest remaining first-order task information. For an infinite-dimensional nuisance tangent space, the orthogonal-projection definition applies directly to its closed span.

Coordinate normalization. Consider separate invertible score transformations

$$\widetilde{s}_{T,e} = U s_{T,e}, \qquad \widetilde{s}_{N,e} = V s_{N,e}.$$

The nuisance span is unchanged, so

$$\widetilde{s}_{T,e}^{\mathrm{eff}} = U s_{T,e}^{\mathrm{eff}}, \qquad \widetilde{F}_e = U F_e U^\top. \tag{A.4}$$

An invertible nuisance-basis change therefore preserves exact task information when the task coordinates remain fixed. Task rescaling transforms information by congruence, preserving rank while generally changing its numerical eigenvalues. Comparisons of λ_e consequently use a common task-coordinate convention.

For one task and one nuisance score with positive raw information, define

$$\rho_e = \frac{J_{TN,e}}{\sqrt{J_{TT,e} J_{NN,e}}}.$$

Then

$$F_e = J_{TT,e}(1 - \rho_e^2), \qquad \frac{F_e}{J_{TT,e}} = 1 - \rho_e^2 \in [0, 1]. \tag{A.5}$$

Nonzero scalar rescaling of either score leaves this fraction unchanged. More generally, for a scalar task, the fraction is invariant under an invertible nuisance-basis change. These identities concern the exact projection; Appendix B specifies the coordinate conventions for its ridge implementation.

The following construction expresses the transverse-information geometry of Mansouri and Arai [2026] in the normalized score coordinates used here.

Proposition A.1 (Task–nuisance tangency)**.** *Let b_e, r_e be centered, orthogonal, unit-norm elements of $L^2(P_e)$. Suppose*

$$\mathcal{T}_{N,e} = \operatorname{span}\{b_e\}$$

and the scalar task score is

$$a_e = b_e + \delta_e r_e, \qquad \delta_e \geq 0.$$

Then

$$J_{TT,e} = 1 + \delta_e^2, \qquad F_e = \delta_e^2, \qquad \frac{F_e}{J_{TT,e}} = \frac{\delta_e^2}{1+\delta_e^2}. \tag{A.6}$$

Proof. The coefficient of the projection of a_e onto b_e is

$$\frac{\langle a_e, b_e\rangle_e}{\|b_e\|_e^2} = 1.$$

Hence its efficient score is $\delta_e r_e$, whose squared norm is δ_e^2. Orthogonality gives $\|a_e\|_e^2 = 1 + \delta_e^2$, and division yields the surviving fraction. □

The construction keeps raw task information at least one while projected information vanishes quadratically. When the nuisance feature supports a predictor with small observed risk, that predictive success coexists with the displayed information collapse. This is the separation used in Section 2.2; Appendix D.3 specifies its synthetic realization.

A.2 Normal Coordinates and the Inverse-Information Tail

We prove the geometric tail law of Mansouri and Arai [2026] under Theorem 1's assumptions, including its asymptotic constant. Let

$$\lambda : \mathbb{R}^d \longrightarrow [0, \infty),$$

and let

$$\mathcal{Z} = \{z : \lambda(z) = 0\}$$

be a nonempty compact C^2 embedded submanifold without boundary, of codimension $c \in \{1, \ldots, d\}$. Choose a tubular radius $r_\star > 0$ and write

$$U_{r_\star}(\mathcal{Z}) = \{z : \operatorname{dist}(z, \mathcal{Z}) < r_\star\}.$$

Assume that λ is continuous on $\overline{U_{r_\star}(\mathcal{Z})}$ and bounded away from zero outside $U_{r_\star}(\mathcal{Z})$.

Let p be the environment density with respect to Lebesgue measure. For $s \in \mathcal{Z}$, let $S(N_s\mathcal{Z})$ denote the unit sphere in its normal space. In normal coordinates $z = s + r\omega$, assume uniformly on the unit normal bundle that

$$\begin{aligned} \lambda(s + r\omega) &= a(s,\omega) r^{2m}(1 + o(1)), \\ p(s + r\omega) &= b(s,\omega) r^{\nu}(1 + o(1)), \end{aligned} \qquad r \downarrow 0, \tag{A.7}$$

where $m > 0$, $\nu > -c$, and a, b are continuous and bounded above and below by positive constants. Define

$$\begin{aligned} \beta &= \frac{c+\nu}{2m}, \\ C_{\mathcal{Z}} &= \frac{1}{c+\nu} \int_{\mathcal{Z}} \int_{S(N_s\mathcal{Z})} b(s,\omega) a(s,\omega)^{-\beta} \, d\sigma_s(\omega) \, d\operatorname{vol}_{\mathcal{Z}}(s), \end{aligned} \tag{A.8}$$

where σ_s is the usual, unnormalized surface measure on the normal unit sphere and $\operatorname{vol}_{\mathcal{Z}}$ is induced volume on $\mathcal{Z}$. In codimension one, the sphere consists of the two normal directions, each with unit mass.

Proof of Theorem 1. First localize the event of small information. For any $r_0 \in (0, r_\star)$, the annulus

$$\{z : r_0 \le \operatorname{dist}(z, \mathcal{Z}) \le r_\star\}$$

is compact and contains no zeros of λ. Continuity gives a positive minimum there. Together with the lower bound outside the outer tube, this implies that

$$\{\lambda < t\} \subset U_{r_0}(\mathcal{Z})$$

for all sufficiently small $t > 0$.

Within the tube, the volume element is

$$\begin{aligned} dz &= j(s, r\omega)\, r^{c-1}\, dr\, d\sigma_s(\omega)\, d\,\mathrm{vol}_{\mathcal{Z}}(s), \\ j(s, r\omega) &= 1 + O(r), \end{aligned} \tag{A.9}$$

uniformly on the unit normal bundle. Fix $\zeta \in (0, 1)$. The uniform expansions and the Jacobian estimate allow r_0 to be chosen so that

$$\begin{aligned} (1-\zeta)a(s,\omega)r^{2m} \le \lambda(s + r\omega) \le (1+\zeta)a(s,\omega)r^{2m}, \\ (1-\zeta)b(s,\omega)r^{\nu} \le p(s + r\omega)j(s, r\omega) \le (1+\zeta)b(s,\omega)r^{\nu} \end{aligned} \tag{A.10}$$

for $0 < r < r_0$.

For sufficiently small t, all relevant radial thresholds lie within this tube. The event $\lambda < t$ contains the radial region

$$r < \left(\frac{t}{(1+\zeta)a(s,\omega)}\right)^{1/(2m)}$$

and is contained in

$$r < \left(\frac{t}{(1-\zeta)a(s,\omega)}\right)^{1/(2m)}.$$

Integrating the corresponding bounds gives

$$\begin{aligned} (1-\zeta)(1+\zeta)^{-\beta} C_{\mathcal{Z}} t^{\beta} &\le \Pr(\lambda < t) \\ &\le (1+\zeta)(1-\zeta)^{-\beta} C_{\mathcal{Z}} t^{\beta}. \end{aligned} \tag{A.11}$$

Here the radial integral is

$$\int_0^{(t/a)^{1/(2m)}} b\, r^{c+\nu-1}\, dr = \frac{b\, a^{-\beta}}{c+\nu}\, t^{\beta}.$$

Letting $t \downarrow 0$ and then $\zeta \downarrow 0$ proves

$$\Pr(\lambda < t) \sim C_{\mathcal{Z}} t^{\beta}.$$

Compactness and the positive bounds on a, b give $0 < C_{\mathcal{Z}} < \infty$. Since $c \ge 1$, the null manifold has zero Lebesgue measure and hence zero environment probability. Taking $t = x^{-1}$ therefore yields

$$\Pr\big(\lambda(z)^{-1} > x\big) \sim C_{\mathcal{Z}} x^{-\beta}. \tag{A.12}$$

This proves Theorem 1. □

Inverse-information moments. For every $p_0 > 0$, the tail-integral identity gives

$$\begin{aligned} \mathbb{E}[\lambda^{-p_0}] &= p_0 \int_0^{\infty} x^{p_0 - 1} \Pr(\lambda^{-1} > x)\, dx, \\ \mathbb{E}[\lambda^{-p_0}] < \infty \quad &\Longleftrightarrow \quad p_0 < \beta. \end{aligned} \tag{A.13}$$

The integral over bounded x is finite, and its tail has the convergence criterion of

$$\int^{\infty} x^{p_0 - \beta - 1}\, dx.$$

In particular, the exact inverse-information mean is finite precisely when $\beta > 1$; at $\beta = 1$, the divergence is logarithmic. Appendix A.4 gives the corresponding certificate limit.

Tangency and independent information branches. For the scalar construction $\lambda = \delta^2$,

$$\Pr(\delta < t) \sim C_\delta t^\gamma$$

immediately gives

$$\Pr(\lambda^{-1} > x) \sim C_\delta x^{-\gamma/2}.$$

The nonnegative residual amplitudes used in the synthetic experiments admit the following scalar portfolio specialization of Mansouri and Arai [2026]. Suppose

$$\lambda_K = \sum_{k=1}^{K} w_k \delta_k^2, \qquad w_k > 0,$$

and $\delta \in [0,\infty)^K$ has a joint density f continuous and positive at the origin from within the orthant. A change of variables gives

$$\begin{aligned} \Pr(\lambda_K < t) &= \frac{t^{K/2}}{\sqrt{\prod_{k=1}^K w_k}} \int_{B_K^+} f\Big(\sqrt{t/w_1}\, u_1, \ldots, \sqrt{t/w_K}\, u_K\Big)\, du \\ &\sim \frac{f(0) v_K}{2^K \sqrt{\prod_{k=1}^K w_k}}\, t^{K/2}, \end{aligned} \tag{A.14}$$

where B_K^+ is the positive-orthant portion of the unit ball and v_K is the volume of the full unit ball. Independence gives

$$f(0) = \prod_{k=1}^{K} f_k(0+).$$

The inverse-information exponent is therefore $K/2$, and its first moment is finite for $K \geq 3$.

This calculation applies to the specified additive scalar information, or to information in a fixed task direction. A minimum-eigenvalue conclusion is governed by Theorem 1 applied to that eigenvalue. Appendix D.3 evaluates the scalar construction.

A.3 Local Classification: Testing Reduction and Tail Aggregation

For a distribution P, let

$$\mathcal{E}_P(g) = R_P(g) - R_P^\star$$

denote population excess classification risk. Fix an environment e and a local family $\mathcal{P}_e^{\mathrm{loc}}$, specified independently of the sample size. Its minimax expected excess risk is

$$\mathfrak{R}_e(n) = \inf_{\widehat{g}_n} \sup_{P \in \mathcal{P}_e^{\mathrm{loc}}} \mathbb{E}_{S_n \sim P^{\otimes n}} \left[\mathcal{E}_P\big(\widehat{g}_n(S_n)\big) \right], \tag{A.15}$$

where the infimum ranges over measurable learning rules. The sample consists of n independent observations from environment e. For randomized rules, the expectation also includes their independent randomization.

Assume $\lambda_e > 0$. For every $0 < t \leq t_{0,e}$, the local family contains $P_{e,+t}$ and $P_{e,-t}$ with task coordinates $\pm t u_e$ along a unit weakest efficient-information direction; nuisance coordinates may vary with the alternative. For constants $K_e, c_e, t_{0,e} > 0$ and $q > 0$, suppose

$$\mathrm{KL}(P_{e,+t} \| P_{e,-t}) \leq K_e t^2 \lambda_e$$

and

$$\inf_g \left\{ \mathcal{E}_{P_{e,+t}}(g) + \mathcal{E}_{P_{e,-t}}(g) \right\} \geq c_e t^q$$

throughout this radius, as in Theorem 2.

Proof of Theorem 2. Fix $0 < t \leq t_{0,e}$ and abbreviate $P_+ = P_{e,+t}$ and $P_- = P_{e,-t}$. The decision-separation assumption gives

$$\mathcal{E}_{P_+}(g) + \mathcal{E}_{P_-}(g) \geq c_e t^q$$

for every classifier g. For an arbitrary learning rule, define the test

$$\psi_t(S_n) = \begin{cases} +, & \mathcal{E}_{P_+}(\widehat{g}_n(S_n)) \le \mathcal{E}_{P_-}(\widehat{g}_n(S_n)), \\ -, & \text{otherwise.} \end{cases} \tag{A.16}$$

If this test selects the wrong alternative, the excess risk under the true alternative is at least $c_e t^q/2$. Thus

$$\begin{aligned} &\max_{\sigma\in\{+,-\}} \mathbb{E}_{P_\sigma^{\otimes n}}[\mathcal{E}_{P_\sigma}(\widehat{g}_n)] \\ &\qquad \ge \frac{c_e t^q}{4}\left\{P_+^{\otimes n}(\psi_t = -) + P_-^{\otimes n}(\psi_t = +)\right\}. \end{aligned} \tag{A.17}$$

The standard two-point testing inequality gives

$$\begin{aligned} &P_+^{\otimes n}(\psi_t = -) + P_-^{\otimes n}(\psi_t = +) \\ &\qquad \ge \frac{1}{2}\exp\left[-\mathrm{KL}\left(P_+^{\otimes n}\|P_-^{\otimes n}\right)\right] \end{aligned} \tag{A.18}$$

[Le Cam, 1986, Tsybakov, 2009]. This inequality follows, for example, from the Hellinger affinity $a(P,Q) = \int \sqrt{pq}$: Jensen's inequality gives

$$a(P,Q) \ge \exp[-\mathrm{KL}(P\|Q)/2],$$

Cauchy–Schwarz gives

$$\mathrm{TV}(P,Q) \le \sqrt{1 - a(P,Q)^2},$$

and the sum of testing errors is at least

$$1 - \mathrm{TV}(P,Q) \ge a(P,Q)^2/2.$$

Independence and the assumed KL bound imply

$$\mathrm{KL}\left(P_+^{\otimes n}\|P_-^{\otimes n}\right) = n\,\mathrm{KL}(P_+\|P_-) \le nK_e t^2 \lambda_e.$$

Both alternatives belong to the full local family. Taking its supremum and then the infimum over learning rules yields, for every admissible t,

$$\mathfrak{R}_e(n) \ge \frac{c_e}{8} t^q \exp(-nK_e t^2 \lambda_e). \tag{A.19}$$

Choose

$$t_n = \min\left\{t_{0,e}, (nK_e\lambda_e)^{-1/2}\right\}.$$

Then $nK_e\lambda_e t_n^2 \le 1$, and hence

$$\mathfrak{R}_e(n) \ge c_0 c_e \min\left\{t_{0,e}^q, (nK_e\lambda_e)^{-q/2}\right\}, \qquad c_0 = \frac{\exp(-1)}{8}. \tag{A.20}$$

This proves Theorem 2. Independent learner randomization can be included in the testing experiment with the same law under both alternatives, leaving the KL calculation unchanged. □

Quadratic decision separation. A sufficient condition for $q = 2$ can be expressed through decision-boundary mass. Suppose the binary alternatives share a feature marginal Q_e, with regression functions

$$\eta_{\pm,t}(x) = P_{e,\pm t}(Y = 1 \mid X = x)$$

and Bayes rules $g^\star_{\pm,t}$. On their disagreement set

$$D_{e,t} = \{x : g^\star_{+,t}(x) \ne g^\star_{-,t}(x)\},$$

any classifier disagrees with at least one Bayes rule. The binary excess-risk identity gives

$$\begin{aligned} &\inf_g \left\{\mathcal{E}_{P_{e,+t}}(g) + \mathcal{E}_{P_{e,-t}}(g)\right\} \\ &\qquad = \int_{D_{e,t}} \min\left\{|2\eta_{+,t}(x) - 1|, |2\eta_{-,t}(x) - 1|\right\} dQ_e(x). \end{aligned} \tag{A.21}$$

Equality holds for unrestricted measurable binary classifiers; the integral is a lower bound for any restricted class.

If, for every $0 < t \le t_{0,e}$, a measurable $A_{e,t} \subseteq D_{e,t}$ satisfies

$$Q_e(A_{e,t}) \ge c_{\mathrm{mass},e}\, t$$

and both margins are at least $c_{\mathrm{margin},e}\, t$ on $A_{e,t}$, for positive constants independent of t, then the right-hand side is at least

$$c_{\mathrm{mass},e}\, c_{\mathrm{margin},e}\, t^2.$$

Thus $q = 2$ and

$$c_e = c_{\mathrm{mass},e}\, c_{\mathrm{margin},e}$$

are valid. This is the moving-boundary regime in which disagreement mass and the relevant margins each scale linearly with the perturbation. Combined with the KL condition of Theorem 2, it gives the inverse-information factor in the quadratic local bound.

Aggregation across environments. Suppose the assumptions hold for Π-almost every environment and the resulting quantities are measurable. Define

$$\mathcal{B}_e(n) = c_e \min\left\{ t_{0,e}^{q}, (nK_e\lambda_e)^{-q/2} \right\}. \tag{A.22}$$

Monotonicity and positive homogeneity of CVaR give

$$\mathrm{CVaR}_{\alpha,e\sim\Pi}(\mathfrak{R}_e(n)) \ge c_0\ \mathrm{CVaR}_{\alpha,e\sim\Pi}(\mathcal{B}_e(n)). \tag{A.23}$$

The environment-specific separation strength, distinguishability constant, and local radius remain inside the aggregation. For $q = 2$, the interior regime

$$nK_e\lambda_e \ge t_{0,e}^{-2}$$

gives $\mathcal{B}_e(n) = c_e/(nK_e\lambda_e)$; the local radius controls the bound beyond that regime. These are the statistical factors retained in Section 3.3 when the inverse-information tail motivates the audit.

A.4 CVaR, Information Coverage, and Stabilization

We use the Fisher-glass certificate of Mansouri and Arai [2026], defined through upper-tail CVaR. Fix $0 < \alpha < 1$. For a nonnegative random variable Z, allowing extended values, upper-tail conditional value-at-risk is

$$\mathrm{CVaR}_\alpha(Z) = \inf_{\tau\in\mathbb{R}} \left\{ \tau + \frac{1}{1-\alpha}\mathbb{E}[(Z-\tau)_+] \right\} \tag{A.24}$$

[Rockafellar and Uryasev, 2000, 2002]. Its upper-tail weighting representation is

$$\begin{aligned} \mathrm{CVaR}_\alpha(Z) &= \sup_{\xi\in\mathcal{W}_\alpha} \mathbb{E}[\xi Z], \\ \mathcal{W}_\alpha &= \left\{ \xi : 0 \le \xi \le \frac{1}{1-\alpha},\ \mathbb{E}\xi = 1 \right\}. \end{aligned} \tag{A.25}$$

Indeed, for every feasible ξ and every τ,

$$\mathbb{E}[\xi Z] = \tau + \mathbb{E}[\xi(Z-\tau)] \le \tau + \frac{\mathbb{E}[(Z-\tau)_+]}{1-\alpha}.$$

Let

$$v_\alpha = \inf\{v : \Pr(Z \le v) \ge \alpha\}.$$

For integrable Z, equality is attained by assigning weight $(1-\alpha)^{-1}$ above v_α, zero below it, and the remaining weight at v_α. Such a choice exists because

$$\Pr(Z > v_\alpha) \le 1-\alpha \le \Pr(Z \ge v_\alpha).$$

Thus atoms at the threshold receive exactly the fraction needed to fill the upper tail. For nonnegative Z with infinite mean, both representations have value $+\infty$.

This representation establishes monotonicity, positive homogeneity, and

$$v_\alpha \le \mathrm{CVaR}_\alpha(Z).$$

For a finite audit with weights $\pi_g \geq 0$ summing to one, it becomes

$$\mathrm{CVaR}_{\alpha,\pi}(Z_g) = \max_{\substack{\sum_g \omega_g = 1 \\ 0 \leq \omega_g \leq \pi_g/(1-\alpha)}} \sum_g \omega_g Z_g. \tag{A.26}$$

These normalized upper-tail weights also provide the convention used in the information-sensitivity calculation of Appendix E.4.

Proof of Proposition 1. Assume $\varepsilon > 0$ and $\lambda_e \in [0, \infty)$ is finite almost surely. Write

$$\begin{aligned} L_e^{(\varepsilon)} &= (\lambda_e + \varepsilon)^{-1}, \\ I_\alpha^{\mathrm{FG}}(\varepsilon) &= \left[\mathrm{CVaR}_{\alpha, e\sim\Pi}\left(L_e^{(\varepsilon)}\right)\right]^{-1}. \end{aligned} \tag{A.27}$$

The explicit argument ε records the stabilization used in Equation (7). If

$$I_\alpha^{\mathrm{FG}}(\varepsilon) \geq \rho > \varepsilon,$$

then

$$v_\alpha(L^{(\varepsilon)}) \leq \mathrm{CVaR}_\alpha(L^{(\varepsilon)}) \leq \rho^{-1}.$$

By the quantile definition,

$$\begin{aligned} \Pr_{e\sim\Pi}(\lambda_e \geq \rho - \varepsilon) &= \Pr_{e\sim\Pi}\left(L_e^{(\varepsilon)} \leq \rho^{-1}\right) \\ &\geq \alpha. \end{aligned} \tag{A.28}$$

This proves Proposition 1, including environments whose information equals the threshold. □

Because $0 < L_e^{(\varepsilon)} \leq \varepsilon^{-1}$, the stabilized certificate satisfies

$$I_\alpha^{\mathrm{FG}}(\varepsilon) \geq \varepsilon.$$

A level $\rho > \varepsilon$ therefore certifies a strictly positive information floor $\rho - \varepsilon$.

Finite-group interpretation. For four equally weighted groups and $\alpha = 0.95$, the group with the largest inverse-information value contains enough probability mass to fill the upper tail. Equation (A.26) gives

$$\begin{aligned} \mathrm{CVaR}_{0.95}(L^{(\varepsilon)}) &= \max_g L_g^{(\varepsilon)}, \\ I_{0.95}^{\mathrm{FG}}(\varepsilon) &= \min_g \lambda_g + \varepsilon. \end{aligned} \tag{A.29}$$

The certificate identifies the weakest absolute information under the declared normalization. The surviving fraction $F_g/J_{TT,g}$ measures relative task–nuisance alignment within a scalar-task group. The two quantities can therefore rank groups differently, as in Section 5.1.

The exact-information limit. Define $L_e^{(0)} = \lambda_e^{-1}$, with $1/0 = +\infty$. As $\varepsilon \downarrow 0$,

$$L_e^{(\varepsilon)} \uparrow L_e^{(0)}.$$

For each feasible ξ, monotone convergence gives

$$\mathbb{E}[\xi L^{(\varepsilon)}] \uparrow \mathbb{E}[\xi L^{(0)}].$$

Taking the supremum over ξ in Equation (A.25) therefore yields

$$\begin{aligned} \mathrm{CVaR}_\alpha(L^{(\varepsilon)}) &\uparrow \mathrm{CVaR}_\alpha(L^{(0)}), \\ I_\alpha^{\mathrm{FG}}(\varepsilon) &\downarrow I_\alpha^{\mathrm{FG}}(0) := \left[\mathrm{CVaR}_\alpha(\lambda^{-1})\right]^{-1}, \end{aligned} \tag{A.30}$$

where $1/(+\infty) = 0$. The interchange follows because the supremum over ξ and the increasing supremum along any sequence $\varepsilon_k \downarrow 0$ commute.

For any nonnegative Z,

$$\mathbb{E}Z \leq \mathrm{CVaR}_\alpha(Z) \leq \frac{\mathbb{E}Z}{1-\alpha}. \tag{A.31}$$

The lower bound uses the feasible weight $\xi = 1$; the upper bound follows from $\tau = 0$ in the variational representation. Hence the exact certificate is strictly positive precisely when

$$\mathbb{E}[\lambda^{-1}] < \infty.$$

Under Theorem 1,

$$\begin{aligned} I_\alpha^{\mathrm{FG}}(0) > 0 \quad &\Longleftrightarrow \quad \beta = \frac{c+\nu}{2m} > 1, \\ I_\alpha^{\mathrm{FG}}(0) = 0 \quad &\text{for } \beta \le 1. \end{aligned} \tag{A.32}$$

This establishes the limiting interpretation in Section 3.3. Information coverage is a population statement under Π; the confidence level of an estimated lower certificate is supplied separately by the estimation analysis. Together, the geometric tail law, the conditional local-risk bound, and the coverage property define the population targets of the operational audit in Section 4.

B Estimation and Certification

This appendix specifies the operational information targets, quantifies ridge bias and sampling error, and constructs a lower certificate for a declared finite audit distribution. The mathematical analysis conditions on the fitted predictor, coordinate maps, calibration, and slice definitions, treating them as fixed independently of the evaluation observations. Appendix C supplies the matrix-estimation argument for learned soft slices evaluated with a frozen gate. Data-dependent normalization in an empirical protocol requires separate treatment; the bounds below do not automatically include its estimation error.

B.1 Operational Models and Matrix Targets

Let P_g denote the evaluation distribution of (X, Y) in slice g. For a fixed predictor with logits $\ell_w(x)$ under the declared calibration convention, introduce

$$\ell_{\theta,\mu}(x) = \ell_w(x) + \Phi_T(x)\theta + \Phi_N(x)\mu, \qquad D(x) = \big(\Phi_T(x)\ \Phi_N(x)\big). \tag{B.1}$$

The columns of Φ_T specify the task contrast, and those of Φ_N specify the nuisance hypothesis. All matrix operations take place in dimension $d = d_T + d_N$. Write $p(x) = \mathrm{softmax}(\ell_w(x))$ and $H(x) = \mathrm{diag}(p(x)) - p(x)p(x)^\top$. Three positive-semidefinite per-example constructions define distinct targets:

$$\begin{aligned} M^{\mathrm{pred}}(x) &= D(x)^\top H(x) D(x), \\ M^{\mathrm{obs}}(x,y) &= s(x,y)s(x,y)^\top, \qquad s(x,y) = D(x)^\top(\mathbf{e}_y - p(x)), \\ M^{\mathrm{feat}}(x) &= v(x)v(x)^\top, \qquad v(x) = (v_T(x)^\top\ v_N(x)^\top)^\top. \end{aligned} \tag{B.2}$$

Here $\mathbf{e}_y$ is the class-indicator vector, and v contains the declared normalized coordinates. Each construction defines $J_g = \mathbb{E}_{P_g}[M]$.

Predictive Fisher. M^{pred} is the likelihood Fisher of the local conditional model with covariate marginal P_g^X and conditional probabilities $\mathrm{softmax}(\ell_{\theta,\mu}(x))$. Averaging its likelihood-score outer product over the fitted conditional label distribution gives M^{pred}. Its Schur complement therefore has the efficient-information interpretation established in Appendix A.1 for that operational model. Predictive Fisher does not require centered features: its likelihood scores have conditional mean zero under the predictive label distribution.

Observed-label score moment. M^{obs} evaluates the score at the observed label. Conditional correct specification within the audited distribution, $P_g(Y = \cdot \mid X) = p(X)$, implies $\mathbb{E}_{P_g}[M^{\mathrm{obs}} \mid X] = M^{\mathrm{pred}}(X)$. More generally, the target is a second moment. Its centered covariance subtracts $\mathbb{E}[s]\mathbb{E}[s]^\top$.

Unweighted feature geometry. M^{feat} measures the second-moment geometry of the chosen coordinates. With $\mu_g = \mathbb{E}_{P_g}[v]$ and $\Sigma_g = \mathbb{E}_{P_g}[(v-\mu_g)(v-\mu_g)^\top]$,

$$J_g = \Sigma_g + \mu_g\mu_g^\top.$$

The original prospective statistic uses J_g; its centered comparison uses Σ_g, estimated with divisor n_g and the same global coordinate scales. This comparison removes within-slice means without rescaling the coordinates.

Normalization is protocol-specific, as recorded in Appendix D; fitted model weights, probes, and temperatures are fixed before final scoring. Some reported score normalizations use the evaluated sample, rather than training moments. Those implementations are not automatically covered by a concentration argument for independent evaluations of a fixed matrix map. The experiments use prespecified nuisance families; Appendix D.5 records the available semantic evidence, fitting protocols, and conditioning audits. Anchor task-preservation tests and formal admission thresholds describe possible extensions of this protocol, not completed admission checks.

Prospective Waterbirds selection uses the uncentered feature–Schur fraction. Appendix D.2 compares it with within-slice covariance and predictive Fisher at the same fixed nuisance ridge. The legacy probability-weighted comparator normalizes $p(1-p)$ weights within each group; this rescales the effective ridge relative to the predictive moment. Hidden-slice discovery uses the predictive moment without that weight normalization and retains same-class shrinkage $\eta = 0.10$ (Appendix C). Removing the nuisance ridge targets the unregularized Schur complement of the declared matrix, including retained shrinkage.

B.2 Ridge Regularization and the Unregularized Target

Partition $J = \left(\begin{smallmatrix} A & B \\ B^\top & C \end{smallmatrix}\right) \succeq 0$, with $C \succ 0$, and define

$$\begin{aligned} S_0(J) &= A - BC^{-1}B^\top, \\ S_{\mathrm{rid}}(J) &= A - B(C + \lambda_{\mathrm{rid}} I)^{-1}B^\top, \qquad \lambda_{\mathrm{rid}} > 0. \end{aligned} \tag{B.3}$$

Both operators are positive semidefinite. The identity $C^{-1} - (C+\lambda_{\mathrm{rid}}I)^{-1} = \lambda_{\mathrm{rid}} C^{-1}(C+\lambda_{\mathrm{rid}}I)^{-1}$ gives the exact gap

$$\Delta_{\mathrm{rid}}(J) := S_{\mathrm{rid}}(J) - S_0(J) = \lambda_{\mathrm{rid}} BC^{-1}(C + \lambda_{\mathrm{rid}} I)^{-1}B^\top \succeq 0. \tag{B.4}$$

The inverse factors commute and have a positive-semidefinite product. If $C \succeq \kappa I$ and $\|B\|_{\mathrm{op}} \le M$, then

$$\|\Delta_{\mathrm{rid}}(J)\|_{\mathrm{op}} \le \frac{\lambda_{\mathrm{rid}} M^2}{\kappa(\kappa + \lambda_{\mathrm{rid}})}. \tag{B.5}$$

The conditioning requirement is substantive. Scalar blocks $A = 1$, $B = \sqrt{c}$, $C = c > 0$ form a positive-semidefinite joint matrix, with $S_0(J) = 0$ but $S_{\mathrm{rid}}(J) = \lambda_{\mathrm{rid}}/(c + \lambda_{\mathrm{rid}})$. The ridge value approaches one as $c/\lambda_{\mathrm{rid}} \downarrow 0$. Nuisance conditioning therefore determines how closely ridge tracks its unregularized target. The isotropic ridge is defined in the declared standardized nuisance coordinates; comparisons retain the same normalization and ridge value.

B.3 Error Allowances and Lower Certificates

Fix G audit groups and weights $\pi_g > 0$ summing to one. Within group g, let $M_{g1}, \dots, M_{gn_g}$ be independent, identically distributed matrices with mean J_g, satisfying $0 \preceq M_{gi} \preceq R_g I_d$ almost surely for a specified finite bound R_g. Set $\widehat{J}_g = n_g^{-1} \sum_i M_{gi}$. Sampling is conditional on the frozen audit specification. The union bound below accommodates dependence between group estimates.

Simultaneous matrix-estimation event. For $\delta \in (0,1)$, define

$$\begin{aligned} u_\delta &= \log \frac{2dG}{\delta}, & t_g &= R_g \left(\sqrt{\frac{2u_\delta}{n_g}} + \frac{2u_\delta}{3n_g} \right), \\ \mathcal{E}_\delta &= \bigcap_{g=1}^{G} \{ \|\widehat{J}_g - J_g\|_{\mathrm{op}} \le t_g \}. \end{aligned} \tag{B.6}$$

Then $\Pr(\mathcal{E}_\delta) \ge 1 - \delta$. Put $X_{gi} = M_{gi} - J_g$. Since M_{gi} and J_g lie between 0 and $R_g I$, $\|X_{gi}\|_{\mathrm{op}} \le R_g$ and $\mathbb{E}[X_{gi}^2] = \mathbb{E}[M_{gi}^2] - J_g^2 \preceq R_g^2 I$. The two-sided self-adjoint matrix Bernstein inequality [Tropp,

2012] yields

$$\Pr(\|\widehat{J}_g - J_g\|_{\mathrm{op}} > t) \le 2d \exp\left(-\frac{n_g t^2}{2R_g^2 + (2/3)R_g t}\right). \tag{B.7}$$

The stated t_g makes this probability at most δ/G; a union bound establishes $\mathcal{E}_\delta$. For score outer products, $\|s\|_2 \le r_g$ supplies $R_g = r_g^2$.

Deterministic Schur perturbation. Let $J, \widehat{J} \succeq 0$ have the same block partition, with $\|\widehat{J} - J\|_{\mathrm{op}} \le t$ and $\|B\|_{\mathrm{op}} \le M$. Then

$$\|S_{\mathrm{rid}}(\widehat{J}) - S_{\mathrm{rid}}(J)\|_{\mathrm{op}} \le h_{\mathrm{rid}}(t, M) := t + \frac{2Mt + t^2}{\lambda_{\mathrm{rid}}} + \frac{M^2 t}{\lambda_{\mathrm{rid}}^2}. \tag{B.8}$$

Set $Q = (C + \lambda_{\mathrm{rid}} I)^{-1}$ and $\widehat{Q} = (\widehat{C} + \lambda_{\mathrm{rid}} I)^{-1}$. Positive semidefiniteness gives $\|Q\|_{\mathrm{op}}, \|\widehat{Q}\|_{\mathrm{op}} \le \lambda_{\mathrm{rid}}^{-1}$, and the resolvent identity gives $\|\widehat{Q} - Q\|_{\mathrm{op}} \le t/\lambda_{\mathrm{rid}}^2$. With $\Delta B = \widehat{B} - B$,

$$\widehat{B}\widehat{Q}\widehat{B}^\top - BQB^\top = \Delta B \widehat{Q}\widehat{B}^\top + B\widehat{Q}\Delta B^\top + B(\widehat{Q} - Q)B^\top. \tag{B.9}$$

The terms have norms at most $t(M + t)/\lambda_{\mathrm{rid}}$, $Mt/\lambda_{\mathrm{rid}}$, and $M^2 t/\lambda_{\mathrm{rid}}^2$. Adding the task-block error proves (B.8).

Allowances from the same event. Assume $C_g \succ 0$. On $\mathcal{E}_\delta$, the observable quantities

$$\overline{M}_g = \|\widehat{B}_g\|_{\mathrm{op}} + t_g, \qquad \underline{\kappa}_g = \lambda_{\min}(\widehat{C}_g) - t_g \tag{B.10}$$

satisfy $\|B_g\|_{\mathrm{op}} \le \overline{M}_g$ and $C_g \succeq \underline{\kappa}_g I$. When every $\underline{\kappa}_g > 0$, set

$$a_g = h_{\mathrm{rid}}(t_g, \overline{M}_g), \qquad d_g = \frac{\lambda_{\mathrm{rid}} \overline{M}_g^2}{\underline{\kappa}_g(\underline{\kappa}_g + \lambda_{\mathrm{rid}})}. \tag{B.11}$$

Here a_g bounds ridge estimation error and d_g bounds the population ridge gap. Both follow from the same simultaneous event.

Lower-certificate guarantee. Let $F_g^0 = S_0(J_g)$, $\lambda_g^0 = \lambda_{\min}(F_g^0)$, and $\widehat{F}_g^{\mathrm{rid}} = S_{\mathrm{rid}}(\widehat{J}_g)$. For $\varepsilon > 0$, define

$$\begin{aligned} \underline{\lambda}_g &= [\lambda_{\min}(\widehat{F}_g^{\mathrm{rid}}) - a_g - d_g]_+, \\ \underline{I}_{\alpha,\pi} &= \left[\mathrm{CVaR}_{\alpha, g\sim\pi}\left((\underline{\lambda}_g + \varepsilon)^{-1}\right)\right]^{-1}. \end{aligned} \tag{B.12}$$

The audit issues this bound when all lower conditioning margins are positive; otherwise it records a conditioning failure. On $\mathcal{E}_\delta$, Weyl's inequality gives

$$\lambda_g^0 \ge \lambda_{\min}(S_{\mathrm{rid}}(J_g)) - d_g \ge \lambda_{\min}(\widehat{F}_g^{\mathrm{rid}}) - a_g - d_g. \tag{B.13}$$

Since $F_g^0 \succeq 0$, clipping preserves $\underline{\lambda}_g \le \lambda_g^0$. Monotonicity of CVaR and reciprocation imply

$$\underline{I}_{\alpha,\pi} \le I_{\alpha,\pi}^0 := \left[\mathrm{CVaR}_{\alpha, g\sim\pi}\left((\lambda_g^0 + \varepsilon)^{-1}\right)\right]^{-1}. \tag{B.14}$$

The probability of issuing an invalid lower certificate is at most δ. On the same event, an issued level $\underline{I}_{\alpha,\pi} \ge \rho > \varepsilon$ implies $\sum_{g:\lambda_g^0 \ge \rho - \varepsilon} \pi_g \ge \alpha$ by Appendix A.4.

Confidence probability concerns held-out observations; information coverage concerns the finite weights π. For a finite, predeclared nuisance family, the union bound can include every group–specification pair. An infimum of certificates preserves coverage separately for each specification. Appendix C supplies the ratio and shrinkage bounds for soft slices used in these deterministic steps.

B.4 Implemented Audit and Numerical Comparison

The reported scalar audits use approximate Gaussian/Wald simulation of the three Fisher moments (A, B, C), with 10^5 draws per environment and nominal 95% confidence. These numerical results use the procedure below. Appendix B.3 supplies a separate finite-sample guarantee under its bounded-matrix assumptions.

Moment covariance and simulation. For score pairs (z_i, b_i) and normalized weights w_i summing to one, put $u_i = (z_i^2, z_i b_i, b_i^2)^\top$. The implementation uses

$$\widehat{m} = \sum_i w_i u_i, \qquad n_{\text{eff}} = \left(\sum_i w_i^2 \right)^{-1}, \\ \widehat{\Sigma}_{\widehat{m}} = \frac{\sum_i w_i^2 (u_i - \widehat{m})(u_i - \widehat{m})^\top}{1 - \sum_i w_i^2}. \tag{B.15}$$

For uniform weights this is the usual sample covariance of u_i divided by n. The covariance is symmetrized before drawing $m^* \sim N(\widehat{m}, \widehat{\Sigma}_{\widehat{m}})$. The simulation holds the realized coordinate and weight conventions fixed; it does not propagate uncertainty from evaluation-fitted normalization. The random-normalization analysis for frozen learned gates is specified separately in Appendix C.

Within a run with G environments, the one-sided Bonferroni tail is $\delta_{\text{W}} = 0.05/(3G)$. Its three essential events concern nuisance conditioning, a ridge-information lower value, and a ridge-gap upper value. Waterbirds uses $G = 4$ and $\delta_{\text{W}} = 0.0041667$; CivilComments uses $G = 18$ and $\delta_{\text{W}} = 0.000925926$. The code requires at least 50 expected draws in this tail. Lower quantiles use NumPy's `method="lower"`; upper gap quantiles use `method="higher"`, without interpolation.

For each draw, ridge and exact Schur values are clipped at zero, with zero assigned when their respective denominators fail the 10^{-12} tolerance. A draw with $C^* \le 10^{-12}$ or $C^* + \lambda_{\text{rid}} \le 10^{-12}$ contributes an infinite gap, rather than being discarded. Otherwise its gap is computed directly as $\lambda_{\text{rid}}(B^*)^2/[C^*(C^* + \lambda_{\text{rid}})]$. This closed form is retained even for a non-positive-semidefinite Gaussian draw; subtracting separately clipped Schur values would understate the gap.

Nuisance, ridge, and direct exact lower quantiles are clipped to $[0, \widehat{C}]$, $[0, \widehat{F}^{\text{rid}}]$, and $[0, \widehat{F}^0]$. Denote them $C_g^L, L_g^{\text{rid}}, L_g^0$, and let U_g^Δ be the upper gap quantile, enlarged to at least the plug-in gap. The implemented lower information is

$$\underline{\lambda}_g^{\text{W}} = \min\{[L_g^{\text{rid}} - U_g^\Delta]_+,\ L_g^0,\ \widehat{F}_g^0\}. \tag{B.16}$$

The direct exact caps only decrease the decomposed lower value and require no extra event for its validity. An infinite upper gap or $C_g^L \le 10^{-12}$ sets the lower value to zero and flags the audit as uncertifiable. Insufficient effective sample size, invalid empirical blocks, and failed numerical identities also prevent a positive certification status.

Calculations use float64, with slack $100\epsilon_{\text{mach}} \max\{|A|, |C|, 1\}^2$ for empirical positive semidefiniteness. Gap identities and certificate ordering use relative tolerance 10^{-10} with scale at least one. A lower certificate at or below $\varepsilon(1+10^{-6})$ is flagged as reaching the numerical floor. Finite-distribution CVaR uses uniform environment weights and includes threshold atoms fractionally. Reported settings are $\lambda_{\text{rid}} = 10^{-3}$, $\varepsilon = 10^{-8}$, and $\alpha = 0.95$. The base audit seed is 20260729; training test audits use 20260730.

Held-out comparisons. Tables B.1–B.2 summarize 75 default-ridge held-out test audits. Parentheses report nominal simultaneous lower conditioning or upper group-gap values within a run. They do not constitute a joint confidence statement across seeds and methods.

Table B.1: Conditioning and ridge gaps. Parentheses contain within-run Wald lower/upper values; entries summarize extrema across runs.

Scope	Audits	Minimum J_{NN}	Maximum group gap
CivilComments diagnostics	5	1.456 (1.417)	0.00116 (0.00133)
Waterbirds diagnostics	5	0.211 (0.178)	0.00125 (0.00166)
Frozen-feature training	30	0.227 (0.192)	0.00144 (0.00199)
End-to-end training	35	0.349 (0.349)	0.00252 (0.00300)
All audits	75	0.211 (0.178)	0.00252 (0.00300)

Every reported run passed the conditioning check and satisfied

$$\underline{I}_{0.95}^{\text{W}} \le \widehat{I}_{0.95}^0 \le \widehat{I}_{0.95}^{\text{rid}}, \tag{B.17}$$

Table B.2: Certificate summaries for the same 75 audits. Inflation is the ridge-minus-unregularized plug-in difference.

Scope	Maximum inflation	Minimum exact plug-in	Minimum Wald lower value
CivilComments diagnostics	0.00018	0.4970	0.4618
Waterbirds diagnostics	0.00071	0.1639	0.1504
Frozen-feature training	0.00062	0.1972	0.1814
End-to-end training	0.00140	0.2185	0.2004
All audits	0.00140	0.1639	0.1504

where W identifies Wald allowances and hats identify plug-in certificates. For the Waterbirds diagnostic these values are 0.1504, 0.1639, and 0.1646. The lower-value caps explicitly enforce the first ordering; numerical ordering does not itself establish confidence coverage.

Training audits use fresh models under the manuscript's frozen-feature and end-to-end protocols. Original training runs did not retain the environment-level blocks needed for this audit, so checkpoint-specific conclusions attach to the fresh runs. Table B.3 gives descriptive five-seed means at manuscript-aligned settings.

Table B.3: Fresh training audits at manuscript-aligned settings: descriptive five-seed means, with selection qualifications below.

Training setting	Ridge plug-in	Exact plug-in	Wald lower value
Frozen-feature Fisher-Glass	0.4002	0.3999	0.3663
Frozen-feature Group DRO + FG	0.6254	0.6252	0.5664
End-to-end Group DRO + FG	0.5447	0.5442	0.4890

The frozen-feature FG row fixes $\lambda_{\mathrm{FG}} = 0.03$. The frozen Group DRO + FG row restricts candidates to $\{0.1, 0.3\}$, then selects the largest validation certificate within two percentage points of the best validation worst-group accuracy in that set. Choices for seeds 17–21 are $(0.3, 0.1, 0.3, 0.3, 0.3)$. The end-to-end row fixes $\lambda_{\mathrm{FG}} = 10^{-4}$. These summaries differ from the full-grid selected runs included in the 75-audit extrema.

With the frozen-feature protocol fixed, runs at ridges $10^{-4}, 10^{-3}, 10^{-2}$ have maximum group gaps 0.0001443, 0.0014374, and 0.0138286, with Wald upper values 0.0001996, 0.0019880, and 0.0191165. These are repeated training-and-audit runs at the stated ridges.

The implementation, report generator, and experiment driver are:

`experiments/ridge_exact_audit.py`,
`experiments/make_ridge_exact_audit_report.py`, and
`experiments/run_w2_ridge_exact_audit.sh`.

Retained run configurations and machine-readable summaries support these numerical values.

C Hidden-Slice Discovery

This appendix specifies discovery and independent evaluation in Section 4.3 and Algorithm 1. The predictor, representation, nuisance coordinate, and matrix construction are fixed for the audit. A learned per-example membership rule is transferred to new observations; the implementation additionally applies a declared support correction to its normalized weights. We distinguish that correction from fitting the gate, and from the fixed-gate ratio target used below. Appendix D.4 reports the populations and controls.

C.1 Gate Family and Support Constraints

Let $\psi(x)$ be the frozen, standardized representation. For task class c, a linear gate assigns membership

$$h_{\phi,c}(x) = \sigma(a_c^\top \psi(x) + b_c), \qquad \sigma(t) = (1 + \exp(-t))^{-1}.$$

The class index and gate specify a class-conditional slice. Task labels remain available throughout discovery and evaluation. On a discovery sample $\mathcal{D}_c$ of size n_c, the normalized membership weights are

$$q_i^0(\phi) = \frac{h_{\phi,c}(x_i)}{\sum_{j \in \mathcal{D}_c} h_{\phi,c}(x_j)}, \qquad i \in \mathcal{D}_c.$$

A general support-constrained family is

$$\mathcal{Q}_c(m, c_q) = \left\{ q \in \mathbb{R}_+^{n_c} : \sum_i q_i = 1, \quad \sum_i q_i^2 \le \frac{1}{m}, \quad \max_i q_i \le \frac{c_q}{m} \right\}, \qquad m > d_T + d_N.$$

The effective sample size is $n_{\text{eff}}(q) = (\sum_i q_i^2)^{-1}$. These constraints limit concentration on a few observations; they do not by themselves guarantee nuisance conditioning.

Implemented mass and support conventions. The Waterbirds target is $\tau_{\text{slice}} = 0.10$ within a task class. During discovery, the sigmoid intercept is solved so that $n_c^{-1} \sum_i h_{\phi,c}(x_i) = \tau_{\text{slice}}$, to numerical tolerance. The saved intercept is the calibrated intercept of the selected checkpoint. Evaluation uses it without re-centering gate scores or recalibrating membership mass.

The code distinguishes the nominal count $m_{\text{nom}} = \tau_{\text{slice}} n_c$ from the ESS floor:

$$m_{\text{ESS}} = \min\{n_c, \max\{1, 0.8 m_{\text{nom}}\}\}, \qquad b_{\max} = \min\{1, 2/m_{\text{nom}}\}.$$

Thus the denominator in the reported $2/m$ cap is the nominal slice size, not the ESS floor. In the unclipped regime, the displayed family represents these settings with $m = m_{\text{ESS}}$ and $c_q = 1.6$.

The function `admissible_weights` first forms $q^0 = h / \sum_i h_i$. If its maximum exceeds $b_{\max}$, it computes the Euclidean projection onto the capped simplex, using 64 bisection iterations and active-set refinement; otherwise the cap step is the identity. Write the result as q^{cap} and $u_i = 1/n_c$. The final weights are $\widetilde{q} = u + t(q^{\text{cap}} - u)$, with the largest $t \in [0, 1]$ satisfying $\sum_i \widetilde{q}_i^2 \le 1/m_{\text{ESS}}$. This radial mixing preserves the cap and unit sum. Discovery and held-out evaluation both apply this support correction using their own class sample size. Consequently, $h_{\phi,c}$ is a frozen per-example function, whereas $\widetilde{q}$ can depend on the full evaluation batch beyond normalization.

Reported held-out membership mass is the mean of raw sigmoid memberships over the selected task class. It is not the fraction of all test images, the fraction of positive weights, or the sum of normalized weights. ESS is computed from $\widetilde{q}$.

Reported slice metrics. For selected-class rows, let l_i be the frozen head's logit, $\widehat{y}_i = \mathbf{1}\{l_i \ge 0\}$, and ℓ_i its binary NLL at temperature one (with the implementation's probability clipping). Reported accuracy, NLL, and post-hoc metadata composition are

$$\text{Acc} = \sum_i \widetilde{q}_i \mathbf{1}\{\widehat{y}_i = y_i\}, \qquad \text{NLL} = \sum_i \widetilde{q}_i \ell_i, \qquad \text{Comp}_g = \sum_i \widetilde{q}_i \mathbf{1}\{G_i = g\}.$$

They use support-corrected weights, not a hard subset, and receive no same-class pooling; shrinkage applies only to the information matrices. Temperature-calibrated NLL is stored separately. The Fisher and high-loss gates use these same conventions. Random subsets use uniform weights on $k = \lceil \tau_{\text{slice}} n_c \rceil$ sampled rows of the Fisher-selected class; the constant-gate comparator uses $1/n_c$. Their performance and information use the same formulas and matrix settings. Unweighted and centered-unweighted controls reevaluate the predictive-Fisher-selected and high-loss-selected gates; they do not rediscover gates with alternative matrix objectives.

For selector distinctness, each frozen gate contributes the $k = \max\{1, \lceil \tau_{\text{slice}} n_c \rceil\}$ test rows with largest raw membership in its selected class. Jaccard is $|A \cap B| / |A \cup B|$ on original row identities, with zero overlap for different selected classes. Selection uses NumPy's `argpartition`, without a secondary tie rule; all ten archived Fisher/high-loss cutoffs are untied. The five-seed mean is 0.0006664, reported as 0.001. Separately, fold stability uses this top-count Jaccard on a common

same-class training-calibration reference sample, averaged over pairs of fold-selected gates. Class averages are weighted by selected-fold counts, excluding classes with fewer than two selected gates, then averaged over seeds. Its reported mean is 0.258. Neither overlap uses positive-membership supports or support-corrected weight ranks.

C.2 Discovery Objective and Shrinkage

Let M_i be the declared per-example matrix from Appendix B.1, and set $\widehat{J}_{\mathrm{pool},c} = n_c^{-1} \sum_{i \in \mathcal{D}_c} M_i$. The constrained information objective is

$$\widehat{J}_{\phi,c}^{(\eta)} = (1-\eta) \sum_{i \in \mathcal{D}_c} q_i(\phi) M_i + \eta \widehat{J}_{\mathrm{pool},c},$$
$$\phi^\star \in \underset{\phi:\, q(\phi) \in \mathcal{Q}_c}{\arg\max} \frac{1}{\lambda_{\min}\left(S_{\lambda_{\mathrm{rid}}}(\widehat{J}_{\phi,c}^{(\eta)})\right) + \varepsilon}, \qquad 0 \le \eta < 1. \tag{C.1}$$

In the implementation, $q(\phi) = \widetilde{q}(\phi)$. A separate search is fitted within each task class; the class and restart with largest discovery utility are selected without evaluation outcomes. The operational geometry is scalar task versus scalar nuisance, so projected information is the scalar Schur complement.

Optimization and numerical settings. The archived protocol uses ten restarts and 1,000 AdamW steps, learning rate 0.03, weight decay 10^{-4}, $\eta = 0.10$, $\lambda_{\mathrm{rid}} = 10^{-3}$, and reciprocal stabilization $\varepsilon = 10^{-8}$. Gate weights start from independent zero-mean Gaussian draws with standard deviation 0.02. The intercept is recalibrated during discovery; AdamW updates the weight vector. The optimized utility is $-\log(\widehat{F}^{\mathrm{rid}} + \varepsilon)$ minus ten times the squared relative membership-mass error. The logarithm preserves the inverse-information ranking at fixed mass. The best feasible finite checkpoint is retained, with the constant gate included as a baseline candidate. The high-loss control uses the same gate architecture, mass calibration, and support restrictions, replacing information utility by weighted prediction loss.

The primary matrix is $M_i = p_i(1-p_i) v_i v_i^\top$, where v_i contains the normalized task and nuisance coordinates and p_i is the temperature-calibrated probability. The factor $p_i(1-p_i)$ is not normalized away. Shrinkage mixes the complete moment block before taking its Schur complement. Its pool is uniform over the same task class in the sample currently evaluated: the discovery fold during optimization, the held-out fold during cross-fitting, and the test class at final evaluation. Evaluation retains $\eta = 0.10$. Unweighted and centered-unweighted calculations are separate robustness targets; the latter additionally estimate weighted coordinate means within the evaluated sample.

Fixed semantic nuisance direction. The CLIP ViT-B/32 coordinate uses the model:

`openai/clip-vit-base-patch32`.

Each prompt embedding is normalized, the five embeddings within a concept are averaged and normalized, and the water-minus-land difference is normalized. Its dot product with the normalized image embedding gives the raw score. The predeclared prompts are:

Water concept	Land concept
a photo with a water background	a photo with a land background
a natural scene with ocean water in the background	a natural scene with forest in the background
a natural scene with lake or river water in the background	a natural scene with grass or ground in the background
a photo taken over open water	a photo taken over dry land
a watery landscape behind the main subject	a terrestrial landscape behind the main subject

The preparation manifest records model revision:

`3d74acf9a28c67741b2f4f2ea7635f0aaf6f0268`.

This direction uses no background or four-way group annotations.

C.3 Cross-Fitting and Final Evaluation

The representation consists of 576 frozen ImageNet MobileNetV3-small features. For each seed, the official training split is stratified by task label into an 85% fitting subset and a 15% calibration subset. The linear head is trained on fitting rows for 1,200 steps with learning rate 0.01 and weight decay 10^{-4}. Feature standardization, task-logit mean and standard deviation, and final nuisance standardization are fitted on that subset and reused unchanged. Preparation also standardizes the raw CLIP score on the official training split; the discovery-stage fit-subset transformation is then applied consistently to every split. Temperature scaling [Guo et al., 2017] minimizes calibration-subset NLL over 400 logarithmically spaced values from 0.05 to 20.

The blind procedure receives images, task labels, split membership, frozen features, and the nuisance coordinate. Background and group annotations remain sealed until all seeds complete certification. The sequence is:

1. **Assess validation transfer.** Five task-label-stratified validation folds rotate complementary discovery and held-out roles, sharing the same assignment across selectors within a seed. Each gate is evaluated on its held-out fold. Pairwise gate stability uses same-class training-calibration reference rows; these comparisons do not refit gates.
2. **Refit the final rule.** With protocol and settings fixed, refit on full validation data for seeds 17–21. Fold fits supply stability and transfer diagnostics and are not ensembled into the final gate.
3. **Freeze fitted objects.** Store the classifier, normalizers, temperature, nuisance direction, gate weight, calibrated intercept, selected class, and support settings before opening the test archive. Membership remains $\sigma(a_c^\top \psi(x) + b_c)$; the declared normalization and support correction of its weight vector use the evaluation batch.
4. **Evaluate independently.** Compute membership, support, moment blocks, and information before revealing metadata. The same-class shrinkage pool is estimated from the test observations.

The matching archived test vectors show inactive support correction for all five primary Fisher gates: stored weights equal normalized raw memberships to numerical precision. It is active for the high-loss control in seed 19, whose ESS rises from 94.48 to the declared floor 102.72. Freezing the membership rule and freezing numerical evaluation weights are therefore different statements. The next subsection gives both the ratio bound and an allowance for the implemented support correction.

C.4 Held-Out Targets and Uncertainty

Condition on all fitting data and fitted objects, and fix the selected class-conditional evaluation distribution P_c. Let h be the resulting per-example gate and $M(Z)$ the frozen per-example matrix map. The unshrunk target is

$$p = \mathbb{E}_{P_c}[h(Z)] > 0, \qquad A = \mathbb{E}_{P_c}[h(Z)M(Z)], \qquad J_h = \frac{A}{p}. \tag{C.2}$$

For n independent evaluation observations from P_c, define

$$\widehat{p} = \frac{1}{n}\sum_{i=1}^{n} h(Z_i), \qquad \widehat{A} = \frac{1}{n}\sum_{i=1}^{n} h(Z_i)M(Z_i), \qquad \widehat{J}_h = \frac{\widehat{A}}{\widehat{p}}.$$

Even with a frozen gate, normalized weights share a random denominator. Concentration must therefore control numerator and membership mass jointly. Here $\widehat{J}_h$ denotes the unprojected ratio; the support-corrected implementation is denoted $\widetilde{J}_h$ below.

Proposition C.1 (Frozen-gate ratio bound)**.** *Assume $0 \le h \le 1$ and $0 \preceq M(Z) \preceq M_{\max} I_d$ almost surely, where finite $M_{\max}$ is specified independently of evaluation observations. For $0 < \delta < 1$, set*

$$u = \log\frac{4d}{\delta}, \qquad r_A = M_{\max}\left(\sqrt{\frac{2u}{n}} + \frac{2u}{3n}\right), \qquad r_p = \sqrt{\frac{\log(4/\delta)}{2n}}.$$

With conditional probability at least $1-\delta$*, both* $\|\widehat{A}-A\|_{\mathrm{op}} \le r_A$ *and* $|\widehat{p}-p| \le r_p$*. On that event, if* $r_p < p$*,*

$$\|\widehat{J}_h - J_h\|_{\mathrm{op}} \le \frac{r_A + \|J_h\|_{\mathrm{op}} r_p}{p - r_p}.$$

Since $\|J_h\|_{\mathrm{op}} \le M_{\max}$*, this is at most* $2(r_A + M_{\max} r_p)/p$ *when* $r_p \le p/2$*. An observable bound, whenever* $\underline{p} = \widehat{p} - r_p > 0$*, is*

$$\|\widehat{J}_h - J_h\|_{\mathrm{op}} \le r_J := \frac{r_A + \|\widehat{J}_h\|_{\mathrm{op}} r_p}{\widehat{p} - r_p}. \tag{C.3}$$

Proof. The centered matrices $h(Z_i)M(Z_i) - A$ have norm at most $M_{\max}$ and variance bounded by $M_{\max}^2 I_d$. Matrix Bernstein gives the matrix event with failure probability at most $\delta/2$ [Tropp, 2012]. Hoeffding's inequality gives the scalar event with failure probability at most $\delta/2$; a union bound gives simultaneous occurrence. The identities

$$\begin{aligned}\widehat{J}_h - J_h &= \frac{(\widehat{A}-A) - J_h(\widehat{p}-p)}{\widehat{p}} \\ &= \frac{(\widehat{A}-A) - \widehat{J}_h(\widehat{p}-p)}{p}\end{aligned}$$

give the bounds using $\widehat{p} \ge p - r_p$ and $p \ge \widehat{p} - r_p$, respectively. Finally, $A \preceq M_{\max} p I_d$, hence $\|J_h\|_{\mathrm{op}} \le M_{\max}$. □

Shrinkage and projected information. When evaluation supplies the same-class pool, set $J_{\mathrm{pool},c} = \mathbb{E}_{P_c}[M]$ and $\widehat{J}_{\mathrm{pool},c} = n^{-1}\sum_i M(Z_i)$. For failure probability δ_{pool}, a matrix Bernstein radius is

$$r_{\mathrm{pool}} = M_{\max}\left[\sqrt{\frac{2\log(2d/\delta_{\mathrm{pool}})}{n}} + \frac{2\log(2d/\delta_{\mathrm{pool}})}{3n}\right].$$

Use δ_{ratio} in Proposition C.1, with $\delta_{\mathrm{ratio}} + \delta_{\mathrm{pool}} = \delta$. On the simultaneous event, whenever $\widehat{p} - r_p > 0$,

$$\begin{aligned}J_h^{(\eta)} &= (1-\eta)J_h + \eta J_{\mathrm{pool},c}, \\ \|\widehat{J}_h^{(\eta)} - J_h^{(\eta)}\|_{\mathrm{op}} &\le (1-\eta)r_J + \eta r_{\mathrm{pool}}.\end{aligned}$$

The estimates share observations; the simultaneous-event argument accommodates their dependence. If the pool is fixed from fitting data, its matrix is part of the conditional target with no additional evaluation error.

Allowance for support correction. Let $q_i^0 = h(Z_i)/\sum_j h(Z_j)$, let $\widetilde{q}$ be the cap/ESS-corrected vector, and set $\widetilde{J}_h = \sum_i \widetilde{q}_i M(Z_i)$. The triangle inequality gives

$$\|\widetilde{J}_h - \widehat{J}_h\|_{\mathrm{op}} \le M_{\max}\sum_i |\widetilde{q}_i - q_i^0| =: r_{\mathrm{proj}}.$$

Consequently, on the same event,

$$\|\widetilde{J}_h^{(\eta)} - J_h^{(\eta)}\|_{\mathrm{op}} \le (1-\eta)(r_J + r_{\mathrm{proj}}) + \eta r_{\mathrm{pool}}. \tag{C.4}$$

This covers batch-dependent support correction without treating it as a fixed per-example transformation. The primary archived point estimates have numerically zero r_{proj}, but that does not imply inactivity on every repeated sample or bootstrap draw.

Appendix B.3 converts these matrix allowances into Schur and eigenvalue allowances for $I_{\alpha,G}$. Issuing a finite-sample lower certificate requires a positive lower membership-mass bound and a positive lower nuisance-conditioning margin. The target retains the declared shrinkage. Evaluation-estimated centering needs additional control of first moments; the proposition concerns a frozen per-example matrix map.

Reported uncertainty. The archived experiment uses 2,000 nonparametric bootstrap replicates within each selected test class. Each replicate resamples aligned membership, task, nuisance, logit, and label rows, recomputes normalized and support-corrected weights, and recomputes the same-class pool. Gate coefficients, intercept, coordinate normalization, and temperature stay fixed. The 95% interval uses the empirical 2.5th and 97.5th percentiles of predictive projected information, with NumPy's default linear quantile interpolation. This is a bootstrap for the implemented estimator, not a computation of the finite-sample radii above.

Across five paired seeds, the interval for $\Delta_s = T_s^{\mathrm{gate}} - T_s^{\mathrm{control}}$ is $\overline{\Delta} \pm t_{0.975,4} s_\Delta / \sqrt{5}$, where s_Δ is the sample standard deviation. Single-procedure intervals apply the same formula to its five values. These intervals summarize variation across seed-specific fitting runs on the shared benchmark; within-seed bootstraps address evaluation-sampling variation conditional on fitted objects. The boundedness and independent-evaluation assumptions of Proposition C.1 supply a separate finite-sample guarantee and are not inferred from bootstrap coverage.

D Experimental Protocols and Additional Results

This appendix gives the protocols and numerical results underlying Section 5. It identifies the operational quantities, records selection and evaluation stages, and presents controls alongside the primary comparisons. Training, repair, and acquisition outcomes appear in Appendix E.

D.1 Experimental Quantities and Aggregation

Operational quantities and data roles. The experiments evaluate complementary information quantities (Table D.1). Prospective selection uses uncentered feature–Schur geometry. Hidden discovery uses temperature-calibrated predictive Fisher with same-class shrinkage $\eta = 0.10$ during discovery and evaluation. Appendix B.1 defines the matrix constructions and normalization conventions.

Table D.1: Diagnostic experiments and replication units.

Experiment	Representation and reported quantity	Seeds
Prospective Waterbirds	Fine-tuned MobileNetV3-small; background-probe surviving fraction	22–26
Synthetic	Normalized task/nuisance coordinates; inverse-information tail and certificate	11–15
Hidden Waterbirds slices	Frozen MobileNetV3-small and fixed CLIP direction; shrunken predictive information	17–21
CivilComments groups	TF-IDF logistic classifier; feature–Schur certificate	13–17
Waterbirds groups	Frozen MobileNetV3-small and logistic fitting; feature–Schur certificate	17–21

Real-data computations use ridge $\lambda_{\mathrm{rid}} = 10^{-3}$ and stabilization $\varepsilon = 10^{-8}$; synthetic computations use $\varepsilon = 10^{-12}$. Tail certificates use $\alpha = 0.95$. Primary task and probe coordinates in the predefined-group diagnostics use training-split normalization. Prospective normalization and the nuisance stress controls have the distinct rules specified below. Prospective and hidden-slice selection rules are frozen before final evaluation.

Uncertainty and replication. Prospective results and multi-seed synthetic summaries use nominal 95% normal-approximation intervals $\bar{z} \pm 1.96 s_z / \sqrt{5}$. Hidden-slice results use two-sided Student-t intervals across five paired seeds; paired contrasts first form within-seed differences. Each hidden-slice test evaluation also includes a bootstrap conditional on the frozen gate, with evaluation weights recomputed as described in Appendix C.

For the original predefined-group aggregates, each seeded run supplies one summary z_j, and the reported nominal 95% normal-approximation interval half-width is

$$h = 1.96 \frac{s_z}{\sqrt{n}}, \qquad s_z^2 = \frac{1}{n-1} \sum_{j=1}^{n} (z_j - \bar{z})^2, \qquad n = 5. \tag{D.1}$$

The seeds are 13–17 for CivilComments and 17–21 for Waterbirds. The implementation returns zero half-width for a single run. These are across-run normal approximations, not resampling intervals over examples or groups. The Waterbirds fitted classifier and primary feature diagnostic are deterministic after frozen-feature extraction and logistic fitting; repeated seeds vary the random controls and reserve anchor. Seed intervals summarize variation with benchmark splits fixed. In the prospective experiment the interval unit is the training seed, not a composite: the 23,176 composites share 5,794 source birds, with two donors per bird and arm.

Appendix E distinguishes other conventions: frozen-feature training uses nominal normal-approximation intervals with 1.96 standard errors, the end-to-end sweep reports standard deviations, and the group-free coefficient comparison uses paired Student-t intervals. Appendix B.4 separately reports the conditional Gaussian/Wald audit. Accuracy changes use percentage points; information changes retain their original units.

D.2 Prospective Nuisance Intervention

Training and frozen selection. ImageNet-initialized MobileNetV3-small models [Russakovsky et al., 2015, Howard et al., 2019] are fully fine-tuned on Waterbirds Sagawa et al. [2020] for 50 epochs, with batch size 32, learning rate 10^{-4}, and bf16 computation on an H100, for seeds 22–26. Validation data fit a temperature and a cross-fitted logistic background probe on the pooled representation. Five probe folds are stratified by validation group; the class-balanced logistic fit allows 3,000 iterations. Features are standardized on validation data. Validation nuisance scores are out of fold and normalized using validation moments; test scores use the full-validation probe and are separately normalized using test moments. Task-logit normalization uses training moments. Test normalization does not feed slice selection. Define

$$\begin{aligned} \mathcal{G}_{\text{eligible}} = \{g :\ & \mathrm{Acc}_{\text{val}}(g) \geq \mathrm{Acc}_{\text{val}}, \\ & \mathrm{NLL}_{\text{val}}(g) \leq \mathrm{median}_h\, \mathrm{NLL}_{\text{val}}(h)\}, \\ g^\star \in \underset{g\in\mathcal{G}_{\text{eligible}}}{\arg\min}\ & \frac{\lambda_{\min}(\widehat{F}_g^{\text{rid}})}{\widehat{J}_{TT,g}}. \end{aligned} \tag{D.2}$$

Temperature scaling [Guo et al., 2017] is fitted once per checkpoint on the full validation split of n_v observations by minimizing mean binary NLL:

$$\begin{aligned} \widehat{T} &= \arg\min_{T\in\mathcal{T}} \frac{1}{n_v}\sum_{i=1}^{n_v} \ell\big(f_w(x_i)/T, y_i\big), \\ \mathcal{T} &= \left\{ \exp\left[\log(0.05) + \frac{j}{399}\log(400)\right] : j = 0,\dots,399 \right\}. \end{aligned} \tag{D.3}$$

Here f_w is the binary logit and ℓ is binary cross-entropy. The implementation clips sigmoid arguments to $[-60, 60]$ and logarithm arguments below at 10^{-12}. It evaluates all 400 logarithmically spaced temperatures in $[0.05, 20]$ and takes the first minimum, so an exact tie favors the smaller temperature. The fitted temperature supplies calibrated probabilities and NLLs; it is frozen before held-out intervention scoring. Eligibility and Table D.2 use uncalibrated NLL; calibrated NLL is stored separately.

Checkpoint hashes, temperature, probe coefficients, slice definition, and audit constants are recorded and hashed before intervention scoring. Evaluation aborts if a checkpoint differs from its recorded value. Tables D.2 and D.3 show validation statistics. Landbird/land and waterbird/water pass the filter; waterbird/water is selected in every seed.

Table D.2: Validation performance and eligibility. Means ± nominal 95% normal-approximation interval half-widths over seeds 22–26; accuracy is a fraction.

Group	n	Accuracy	NLL	Eligible
Landbird/land	467	0.990 ± 0.005	0.038 ± 0.016	Yes
Landbird/water	466	0.703 ± 0.019	1.670 ± 0.156	No
Waterbird/land	133	0.390 ± 0.021	7.718 ± 0.800	No
Waterbird/water	133	0.920 ± 0.013	0.638 ± 0.038	Yes

Table D.3: Validation information quantities from uncentered feature moments; the same aggregation as Table D.2.

Group	Raw information	Projected information	Surviving fraction
Landbird/land	0.737 ± 0.046	0.436 ± 0.045	0.590 ± 0.027
Landbird/water	0.794 ± 0.032	0.289 ± 0.014	0.365 ± 0.026
Waterbird/land	0.973 ± 0.033	0.764 ± 0.031	0.786 ± 0.036
Waterbird/water	2.063 ± 0.054	0.511 ± 0.021	0.248 ± 0.015

Intervention construction. Waterbirds images combine CUB birds Wah et al. [2011] and Places backgrounds Zhou et al. [2018] under soft CUB mattes. For each held-out bird, its own background is recovered verbatim outside the dilated foreground mask and Telea inpainting Telea [2004] is used inside it. The extracted bird layer is recomposed onto a donor background of the same or opposite type. The same-type arm undergoes the same recomposition and supplies the artifact control. Inpainting uses radius 5 and mask dilation 2. Donors are training images with foreground fraction at most 0.08. For each draw the deterministic candidate sequence considers at most four donors, taking the first with visibly inpainted area at most 0.03, or the least-inpainted of four if none qualifies.

Two donors per bird and arm produce 23,176 composites from 5,794 held-out birds. Visibly inpainted area has mean 4.13%, 95th percentile 8.45%, and maximum 20.37%; arm means are 4.13% and 4.12%. Both arms use the same extracted foreground layer, matte, and label. Fully opaque foreground pixels are unchanged before JPEG encoding; soft-edge mixtures change with the background. Composites are stored as JPEG at quality 95 with chroma subsampling disabled. All four groups are generated, and both arms evaluate the same source birds.

Held-out outcomes and confidence-conditioned flips. The controlled drop is same-type minus opposite-type accuracy (Table D.4). Correctness is averaged over two donor draws per bird before averaging over birds. Among originally correct predictions in the selected slice, 56.4% flip under opposite-type backgrounds versus 3.3% under the same-type control. Within the most confident half of correct predictions, the rates are 41.8% and 0.7%. Confidence is calibrated true-label probability; the half-sample rule uses the within-slice median among originally correct predictions and includes ties. The fixed threshold $p \geq 0.9$ selects none on this slice. Table D.5 gives all-group results.

Table D.4: Held-out intervention outcomes. Accuracies are seed means in percent. Controlled drops are percentage points with nominal 95% normal-approximation intervals over seeds 22–26.

Group	Birds	Original	Same type	Opposite type	Controlled drop
Landbird/land	2,255	98.9	99.3	72.6	26.7 ± 1.8
Landbird/water	2,255	71.3	72.3	99.2	-26.9 ± 2.0
Waterbird/land	642	42.3	40.6	92.3	-51.7 ± 2.0
Waterbird/water	642	**92.0**	**93.2**	**40.6**	$\mathbf{52.6 \pm 1.4}$

Table D.5: Opposite-type flip rates among originally correct predictions. Rates are seed means in percent; counts are reported mean confident-half counts.

Group	All originally correct	Confident half	Confident-half count
Landbird/land	26.8	14.8	1,122
Landbird/water	0.2	0.1	805
Waterbird/land	1.2	0.3	136
Waterbird/water	56.4	41.8	299

Selection comparisons. Table D.6 compares selectors within their stated candidate sets. Reapplying the original eligibility rule to the archived validation records leaves landbird/land and waterbird/water in each seed. Within this set, minimum validation accuracy and maximum uncalibrated validation NLL are evaluated separately and agree with the original surviving-fraction selector in all five seeds. The all-group comparisons answer a different selection question. Landbird/water determines the model-level certificate.

Table D.6: Selection comparisons. Eligibility and selection are evaluated separately within each training seed. "Eligible" uses Equation (D.2); drops are seed means in percentage points.

Statistic	Candidates	Selected group, seeds	Drop
Minimum raw task information	Eligible	Landbird/land, 5/5	26.7
Minimum absolute projected information	Eligible	Landbird/land, 5/5	26.7
Minimum surviving fraction	Eligible	Waterbird/water, 5/5	52.6
Minimum validation accuracy or maximum validation NLL	Eligible	Waterbird/water, 5/5	52.6
Minimum normalized-weight fraction	Eligible	Waterbird/water, 5/5	52.6
Maximum NLL or minimum accuracy	All four	Waterbird/land, 5/5	−51.7
Group determining $I^{\mathrm{FG}}_{0.95}$	All four	Landbird/water, 5/5	−26.9

Centering and predictive-Fisher checks. This post hoc reanalysis retains the five frozen checkpoints, validation coordinates $v_i = (z_i, b_i)^\top$, global scales, temperatures, original eligible sets, and existing intervention outcomes. No model or probe is refitted. Task coordinate z retains training-logit normalization; nuisance coordinate b retains the global validation normalization of the out-of-fold probe. Write $\langle\cdot\rangle_g = n_g^{-1}\sum_{i\in g}(\cdot)$ and compare

$$J_g = \langle vv^\top\rangle_g, \qquad \Sigma_g = \langle (v-\mu_g)(v-\mu_g)^\top\rangle_g, \qquad \mu_g = \langle v\rangle_g.$$

Centering uses divisor n_g and does not rescale the coordinates. For the predictive check, specify the local Bernoulli model

$$\operatorname{logit} p_{\theta,\xi}(x) = f_w(x)/\widehat{T} + \theta z(x) + \xi b(x), \qquad (\theta,\xi) = (0,0).$$

Its likelihood score is $(Y - p(x))v(x)$, with conditional predictive mean zero even when v has nonzero slice mean. The resulting predictive Fisher and common surviving fraction are

$$I_g = \langle qvv^\top\rangle_g, \quad q = p(1-p), \qquad r_\rho(A) = 1 - \frac{A_{zb}^2}{A_{zz}(A_{bb}+\rho)}, \quad \rho = 10^{-3}.$$

The perturbations are applied after temperature calibration, and only the declared background coordinate is profiled. The archived probability-weighted comparator divides weights by their within-group sum, giving $K_g = I_g/\bar{q}_g$, where $\bar{q}_g = \langle q\rangle_g$. Thus $r_\rho(K_g) = r_{\bar{q}_g\rho}(I_g)$: normalization preserves the unregularized fraction but changes the effective ridge when its numerical value is fixed. We reproduce the archived comparator separately from the predictive Fisher with fixed ridge.

Table D.7: Frozen-coordinate sensitivity within the original eligible set, evaluated separately in each seed. LL denotes landbird/land and WW waterbird/water. Fractions and controlled drops are means with nominal 95% normal-approximation intervals (±1.96 standard errors) over five seeds; drops use the existing interventions and are in percentage points.

Construction	LL fraction	WW fraction	Selected, seeds	Drop
Uncentered J_g	0.590 ± 0.027	0.248 ± 0.015	WW, 5/5	52.6 ± 1.4
Centered Σ_g	0.779 ± 0.028	0.855 ± 0.032	LL, 5/5	26.7 ± 1.8
Predictive Fisher I_g	0.987 ± 0.014	0.273 ± 0.016	WW, 5/5	52.6 ± 1.4
Normalized weights K_g	0.987 ± 0.014	0.269 ± 0.016	WW, 5/5	52.6 ± 1.4

Centering selects landbird/land in all five seeds, while predictive Fisher retains waterbird/water (Table D.7). The original ranking therefore depends in part on slice means. For example, for waterbird/water in seed 22, $\mu_z^2 = 2.094$ and $\Sigma_{zz} = 0.0636$, so most of the uncentered task moment is contributed by the squared mean. Table D.8 records the moments and fractions; full-precision matrices, weights, per-seed selections, matched outcomes, and input hashes accompany the reproducible reanalysis.

Individual flip prediction. We predict whether an originally correct prediction flips under opposite-type intervention, retaining two donor rows per bird. The baseline uses uncalibrated NLL, calibrated true-label confidence, absolute raw logit (margin), and standardized task logit z; the augmented design adds $\beta_g b$ and $|z - \beta_g b|$, where $\beta_g = \widehat{J}_{TN,g}/(\widehat{J}_{NN,g} + 10^{-3})$ is frozen from the validation uncentered

Table D.8: Per-seed validation means, covariance entries, and fractions, using the original global scales. LL and WW are defined in Table D.7. Values are rounded; $J_g = \Sigma_g + \mu_g \mu_g^\top$. The selected group minimizes the fraction within each seed's two eligible groups.

Seed	Group	μ_z	μ_b	Σ_{zz}	Σ_{zb}	Σ_{bb}	$r_\rho(J_g)$	$r_\rho(\Sigma_g)$	$r_\rho(I_g)$
22	LL	−0.412	−0.840	0.501	0.229	0.407	0.5570	0.7430	0.9817
22	WW	1.447	0.812	0.064	0.047	0.238	0.2291	0.8542	0.2587
23	LL	−0.433	−0.841	0.551	0.214	0.434	0.6048	0.8098	0.9920
23	WW	1.416	0.787	0.068	0.051	0.253	0.2494	0.8481	0.2729
24	LL	−0.419	−0.838	0.626	0.217	0.403	0.6356	0.8132	0.9620
24	WW	1.388	0.772	0.066	0.055	0.283	0.2755	0.8359	0.3032
25	LL	−0.420	−0.822	0.596	0.274	0.505	0.5803	0.7514	0.9998
25	WW	1.414	0.792	0.069	0.036	0.221	0.2392	0.9156	0.2571
26	LL	−0.431	−0.841	0.516	0.220	0.420	0.5716	0.7779	0.9998
26	WW	1.398	0.864	0.068	0.062	0.310	0.2460	0.8189	0.2732

block. The auxiliary predictor is logistic regression with intercept and ℓ_2 penalty (scikit-learn 1.6.1: $C = 1$, L-BFGS, tolerance 10^{-4}, at most 5,000 iterations, random state 0, no class weighting). Five nonshuffled CUB-species `GroupKFold` folds keep each species and both donor rows together, using identical folds for both predictors. Every auxiliary design column, including group indicators when present, is centered and scaled using only training-fold means and population standard deviations; standard deviations below 10^{-9} are replaced by one. The fixed upstream z, b normalization is as specified above. There is no hyperparameter tuning or inner cross-validation. For seed s, with N_s donor rows, flip indicators y_{si}, and out-of-fold probabilities q_{si}^A, q_{si}^B for augmented and baseline predictors, define

$$G_s = \frac{1}{N_s} \sum_{i=1}^{N_s} \left[\mathrm{LL}(y_{si}, q_{si}^A) - \mathrm{LL}(y_{si}, q_{si}^B) \right], \qquad \mathrm{LL}(y, q) = y \log q + (1 - y) \log(1 - q). \tag{D.4}$$

Probabilities are clipped to $[10^{-12}, 1 - 10^{-12}]$; G_s is a natural-log gain in nats per donor row. Predictions from all five held-out folds are pooled within each seed before computing log likelihood, average precision (AUPRC), and AUROC. Table D.9 averages within-seed paired changes equally over seeds 22–26; counts and flip rates are also seed means. The added coordinates improve prediction across groups without indicators and within the selected slice. With group indicators, AUPRC changes by -0.007 and AUROC by $+0.003$. This comparison measures the incremental predictive value of the nuisance-related coordinates; attribution to the scalar information statistic remains unresolved. Recovered inputs, original code, paired seed results, and the executable `refit.py` and `aggregate.py` commands are provided under `reanalysis/flip_prediction/`.

Table D.9: Incremental flip prediction. Parentheses count seeds with improvement; changes are paired within seed.

Analysis	Rows	Flip rate	Log-likelihood gain	ΔAUPRC	ΔAUROC
All groups, no indicators	9,398	0.199	+0.016	+0.035 (5/5)	+0.096 (5/5)
All groups, with indicators	9,398	0.199	+0.001	−0.007 (0/5)	+0.003 (5/5)
Selected slice	1,181	0.564	+0.013	+0.026 (5/5)	+0.015 (5/5)

Development and run history. The surviving-fraction rule was adopted after a pilot on seeds 17–21. Its absolute-information rule selected landbird/land, whose accuracy changed from 99.2% under same-type intervention to 70.0% under opposite-type intervention. Among originally correct predictions with calibrated confidence $p \geq 0.9$, flip rates were 22.2% and 0.3% for opposite-type and control arms.

The revised rule was frozen for newly trained seeds 22–26 on the same benchmark. Documented reruns with the same code, seeds, and rules give a pilot drop of 29.2 points versus the original 28.9, and a subsequent-run drop of 52.6 versus 52.7 points. The latter opposite-type accuracies are 40.6% and 40.3%. Tables here consistently use the current rerun. The experiment measures sensitivity along

the background attribute encoded by the probe. Figure D.1 assembles selection and outcomes. The Fisher-Glass training arm uses the same construction; Appendix E.3 reports its coefficient selection and behavioral outcomes.

D.3 Synthetic Classification and Tail Checks

Generator and measured quantities. For each environment draw $U_e \sim \mathrm{Uniform}(0,1)$ and set $\delta_e = \delta_{\max} U_e^{1/\gamma}$. Conditional on e, each observation draws independent standard normal s and r, defines task coordinate $z = s + \delta_e r$ and label $Y = \mathbf{1}\{z > 0\}$, and is scored by nuisance-only logit $4s$. Thus

$$\Pr(\delta_e < t) = \left(\frac{t}{\delta_{\max}}\right)^{\gamma} \quad (0 \le t \le \delta_{\max}), \qquad J_{TT,e} = 1 + \delta_e^2, \qquad F_e^0 = \delta_e^2. \tag{D.5}$$

The Gaussian coordinates give population second moments exactly; classifier loss and accuracy use 256 independent observations per environment. The reported $\gamma = 1$ runs use $\delta_{\max} = 0.35$ and 300,000 environments. Seeds 11–15 give Table D.10. The exponent is fitted by log-survival regression over the largest 5% of inverse-information values. Empirical CVaR averages values at or above the empirical α-quantile.

Table D.10: Synthetic summaries over seeds 11–15. Means ± nominal 95% normal-approximation interval half-widths; zero half-widths reflect reported rounding.

Quantity	Estimate
Mean accuracy	$94.54 \pm 0.00\%$
Fifth-percentile environment accuracy	$88.67 \pm 0.00\%$
Loss-$\mathrm{CVaR}_{0.95}$	0.262 ± 0.000
$I^{\mathrm{FG}}_{0.95}$	$(1.22 \pm 0.53) \times 10^{-8}$
Inverse-information exponent $\widehat{\beta}$	0.498 ± 0.003

Figure 2 regenerates the initial environment draw from documented seed 11. It uses $\delta_e \sim \mathrm{Uniform}(0, 0.35)$ and $L_e = (\delta_e^2 + 10^{-12})^{-1}$. This is a regeneration from the documented generator, not recovered raw output. Its survival curve and the five-seed exponent summarize different replication levels.

Structural comparisons. Table D.11 separates the reported single-seed mass sweep from the five-seed independent-branch experiment. These numerically check the geometric tail law in Theorem 1 and the scalar portfolio law of Mansouri and Arai [2026], respectively. For K branches, independent draws $\delta_{e,1}, \ldots, \delta_{e,K}$ give projected information $K^{-1}\sum_{j=1}^{K} \delta_{e,j}^2$. Equal positive weights change the tail constant but not the exponent. The mass sweep follows $\beta = \gamma/2$; at $\gamma = 1$, independent branches follow $\beta = K/2$.

Table D.11: Structural tail comparisons. Mass-sweep estimates are reported single-seed values; branch estimates are five-seed means ± nominal 95% normal-approximation interval half-widths.

Construction	Setting	Predicted β	Estimated β
Mass near the null	$\gamma = 1$	0.5	0.509
Mass near the null	$\gamma = 2$	1	0.985
Mass near the null	$\gamma = 4$	2	2.001
Independent branches	$K = 1$	0.5	0.499 ± 0.003
Independent branches	$K = 2$	1	1.000 ± 0.009
Independent branches	$K = 3$	1.5	1.508 ± 0.015

The reserve experiment checks the Fisher-reserve principle of Mansouri and Arai [2026]. An anchor with residual amplitude $\delta_0 = 0.5$ supplies information $\rho_0 = 0.25$. Reserve mass $r = 0.05$ produces mixed information $(1-r)\delta_e^2 + r\rho_0$, floor $r\rho_0 = 0.0125$, and measured certificate 0.01260 ± 0.000001 across five seeds. Figure D.2 displays both exponent comparisons and this evaluated point; the reserve line is the analytic floor. Appendix E.1 gives the guarantee.

D.4 Hidden-Slice Discovery Results

Information available to discovery. The blind procedure receives images, task labels, official split membership, frozen ImageNet MobileNetV3-small features Howard et al. [2019], and a linear head trained on the official training split. Its nuisance coordinate is the image cosine score along a fixed CLIP ViT-B/32 Radford et al. [2021] water-background-minus-land-background text direction from a predeclared prompt ensemble. Background and four-way group annotations remain sealed until all seeds complete certification, then enter post hoc analysis.

Appendix C specifies class-conditional gates, support constraints, shrinkage, and five-fold cross-fitting. The final gate is refitted on full validation data, frozen, and applied to the untouched test split for seeds 17–21. The gate is fixed, while the prescribed admissibility transformation of its weights is recomputed on each evaluation sample; Appendix C distinguishes this from a simple fixed-gate ratio.

The numerical cohort comes from `waterbirds_hidden_slice_v2_corrected_results.tar.gz`. The public release provides its numerical records under `artifacts/hidden_slice_corrected_v2/`, with original archive/member hashes in `ARTIFACT_MANIFEST.json`. The retained `main_table.csv`, `aggregate.json`, and per-seed records support the tables below. This is the corrected v2 cohort; the older `waterbirds_hidden_slice_v2_results_full.tar.gz` contains different results and is excluded from the release.

Reported slice metrics. Accuracy, uncalibrated NLL, and metadata composition use normalized support-corrected membership weights on the selected task class, without the same-class pooling used for information. The primary five CLIP gates need no support correction, so those weights equal normalized raw membership weights there. The high-loss and random selectors have the same target within-class discovery mass 0.10; held-out gate mass can drift. Appendix C.1 defines these conventions for each control and gives the raw-membership top-count Jaccard rule, its treatment of ties, and the separate fold-overlap statistic.

Selector comparisons. All five selectors use the same fitted classifiers, with average accuracy $76.3 \pm 0.7\%$ and post hoc oracle worst-group accuracy $57.3 \pm 1.3\%$. The CLIP search finds a slice with 98.8% accuracy and Jaccard overlap 0.001 with the high-loss slice. Table D.12 reports the information/loss comparison; Table D.13 gives paired contrasts.

Table D.12: Held-out hidden-slice selectors. Means $\pm$ 95% Student-t interval half-widths over five paired seeds; predictive information retains calibration and same-class shrinkage $\eta = 0.10$.

Selector	Predictive projected information	Slice NLL
Fisher search, fixed CLIP direction	0.005 ± 0.001	0.031 ± 0.017
High loss, same target within-class mass	0.019 ± 0.006	5.779 ± 0.496
Random, same target within-class mass	0.013 ± 0.001	1.025 ± 0.059
Constant class-conditional gate	0.013 ± 0.001	1.027 ± 0.049
Fisher search, random direction	0.006 ± 0.001	0.031 ± 0.015

Table D.13: CLIP-direction Fisher-search information minus the comparator. Negative values indicate lower information; intervals are paired 95% Student-t intervals.

Information	Comparison	Difference	95% interval
Predictive	High-loss slice	-0.014	$[-0.020, -0.008]$
Predictive	Random slice	-0.008	$[-0.009, -0.007]$
Predictive	Constant gate	-0.008	$[-0.009, -0.007]$
Predictive	Random direction	-0.0004	$[-0.0015, 0.0007]$
Unweighted feature	High-loss slice	-0.034	$[-0.106, 0.039]$
Centered unweighted	High-loss slice	-0.071	$[-0.166, 0.024]$

The CLIP Fisher, random-direction Fisher, random-slice, and constant-gate selectors use landbirds in all five seeds. High-loss selection uses landbirds in seeds 17, 18, and 21, and waterbirds in seeds 19

and 20. The latter two comparisons have zero Jaccard overlap with the CLIP slice by construction; the three same-class overlaps are 0.00222, 0.00111, and 0. The mean 0.000666 (reported as 0.001) therefore combines class-selection differences and within-class population differences.

Predictive information is lower than for high-loss, random, and constant-gate selection. The unweighted and centered-unweighted rows reevaluate the frozen gates selected by the predictive-Fisher and high-loss procedures; they do not rediscover gates under alternative matrix objectives. The random-direction contrast and both feature-geometry intervals include zero. Figure D.3 displays these comparisons on separate scales.

Support, stability, and sensitivity. Gate-score correlation and top-slice overlap are evaluated on the common same-class training-calibration reference set. Across validation folds, their values are $0.419\,[0.353, 0.485]$ and $0.258\,[0.225, 0.291]$, respectively. Held-out effective sample size is $414\,[383, 445]$. Reported mass $0.083\,[0.077, 0.089]$ is mean raw sigmoid membership *within the selected task class*, against target 0.10; it is not the fraction of the full test set. Effective sample size uses normalized admissible weights. The primary CLIP gate selects the landbird class in all seeds, containing 4,510 test examples.

The sensitivity sweeps use validation cross-fitting, not the final test evaluation. On mass grid $\{0.05, 0.10, 0.20\}$, Fisher-selector information ranges from 0.004 to 0.005, versus 0.012–0.027 for high-loss selection. On shrinkage grid $\{0.05, 0.10, 0.20\}$, its information ranges from 0.004 to 0.006.

Post-certification metadata identify the CLIP slice as 97.0% landbird/land, the random-direction slice as 94.4% landbird/land, and the high-loss slice as 58.4% landbird/water. CLIP-slice predictive information is below the weakest predefined group's in every seed. An oracle sensitivity analysis using a metadata-trained background probe after blind freeze gives projected information 0.003 ± 0.000, NLL 0.009 ± 0.004, and 99.1% landbird/land composition. The information half-width rounds to zero at the shown precision. These comparisons locate the population and quantify nuisance-coordinate sensitivity; matched-continuation and repair results appear in Appendix E.3.

D.5 Predefined-Group Diagnostics

Models and fitting. CivilComments-WILDS [Borkan et al., 2019, Koh et al., 2021] uses TF-IDF unigram/bigram features (50,000 maximum features, minimum document frequency 5, sublinear term frequency) and class-balanced stochastic-gradient logistic fitting with ℓ_2 penalty coefficient 10^{-5}, at most 50 passes, and tolerance 10^{-3}. A toxicity model supplies the task score; an identity-any probe supplies the nuisance score. Identity-by-label groups with fewer than 100 examples are excluded. The single run uses seed 13; the aggregate uses seeds 13–17. The diagnostic fits log survival over the largest 50% of group inverse-information values, requiring at least eight groups and four fitted tail points.

Waterbirds uses the Waterbirds loader supplied by the WILDS package [Koh et al., 2021], frozen ImageNet MobileNetV3-small features Howard et al. [2019], extraction batch size 64, and class-balanced logistic fitting by L-BFGS with at most 1,000 iterations for task and background models. It reports four group information values directly; no tail exponent is fitted to four groups. The classifier and primary diagnostic remain fixed across repeated seeds, while random controls and the reserve anchor vary. Primary task/probe scores and metadata nuisance scores use training-moment standardization.

Table D.14: Original predefined-group diagnostics. Means $\pm$ nominal 95% normal-approximation interval half-widths $1.96 s_z/\sqrt{5}$ over seeded runs; Waterbirds zero half-widths reflect its fixed-model protocol. These summaries are separate from fresh audit replications in Appendix B.4.

Dataset	Average accuracy (%)	Worst-group accuracy (%)	Loss-$\mathrm{CVaR}_{0.95}$	$I^{\mathrm{FG}}_{0.95}$
CivilComments	86.10 ± 0.22	43.75 ± 0.64	0.998 ± 0.012	0.500 ± 0.001
Waterbirds	77.11 ± 0.00	56.54 ± 0.00	1.741 ± 0.000	0.165 ± 0.000

For the fixed CivilComments diagnostic, the weakest-information group is `muslim`/$y = 0$ with 60.1% accuracy, while `black`/$y = 0$ has the lowest accuracy, 44.0%. For Waterbirds, landbird/land is weakest in information with 97.3% accuracy; waterbird/land has lowest accuracy, 56.5%. Its

reported inverse-information 95th percentile is 5.90. Information and accuracy rankings identify different populations in both datasets.

Nuisance-specification sensitivity. Table D.16 holds each fitted classifier and its groups fixed while varying the nuisance coordinate. The reported robust family is the prespecified semantic family $\mathcal{B}_{\text{sem}} = \{b_{\text{meta}}, b_{\text{probe}}\}$: the standardized binary `identity_any` annotation and its fitted identity probe on CivilComments, or the standardized background annotation and its fitted background probe on Waterbirds. Both probes are trained only on the official training split, without cross-fitting, with the fitting settings above. The family minimum is 0.499 on CivilComments and 0.165 on Waterbirds. Remaining rows are stress controls. Table D.15 records the evidence and the checks not performed; membership in this family does not assert passage of a formal task-preservation admission test.

Table D.15: Documented specification procedure for the predefined-group nuisance family.

Check	Evidence and status
Semantic relevance	Metadata directly names identity presence or background; supervised probes fit those annotations. No held-out relevance threshold selects family members.
Task preservation	No anchor-distribution preservation statistic, acceptance tolerance, or pass/fail outcome is documented.
Fold stability	No fold-based admission screen is performed for these two-member diagnostic families. Hidden-gate fold stability in Appendix D.4 is a separate analysis.
Conditioning	A 10^{-3} nuisance ridge is used; no nuisance-block conditioning threshold admits or rejects a specification. Fresh exact/ridge diagnostics in Appendix B.4 form a separate cohort.

The reserve anchors used elsewhere are construction inputs, not task-preservation admission samples: CivilComments balances training examples with `identity_any`$= 1$ across task labels, while Waterbirds balances training examples across its four label/background groups. No task-preservation acceptance rule was applied to those anchors.

Table D.16: Nuisance sensitivity in the original fixed-model diagnostics. Entries are $I^{\text{FG}}_{0.95}$; worst-group accuracy stays at 44.0% on CivilComments and 56.5% on Waterbirds.

Specification	CivilComments	Waterbirds	Role
Metadata	0.587	0.266	Semantic-family member
Learned probe	0.499	0.165	Semantic-family minimum
Random	0.588	0.482	Negative control
Under-projected	0.588	0.482	Under-projection control
Over-projected	0.001	0.012	Task-erasure control
Hidden-loss	0.060	0.177	Loss-coordinate stress control

The random control is an independent standard-normal coordinate; under-projection sets the nuisance score to zero. Over-projection uses the standardized task score plus Gaussian noise of standard deviation 0.01, deliberately erasing task information. The implementation's "hidden-loss" coordinate is observed test loss, with original groups unchanged. Random, over-projected, and loss coordinates are standardized on their evaluated arrays. They are diagnostic controls, not independently learned nuisance scores. CivilComments metadata and random certificates are nearly equal (0.587 versus 0.588), whereas Waterbirds probe and random values differ (0.165 versus 0.482), exposing dependence on the specification.

The implementations are:

`experiments/civilcomments_fisher_glass.py`,
`experiments/waterbirds_fisher_glass.py`,
`experiments/synthetic_fisher_glass.py`, and
`experiments/multiseed_fisher_glass_diagnostics.py`.

Original diagnostic runs used an Intel i9-11900F workstation with 32 GB RAM. Reported times were approximately one minute for CivilComments and three minutes for the first Waterbirds extraction/fitting run, with later runs reusing cached features.

Figures D.1–D.3 show the prospective intervention controls, synthetic structural checks, and paired discovery contrasts.

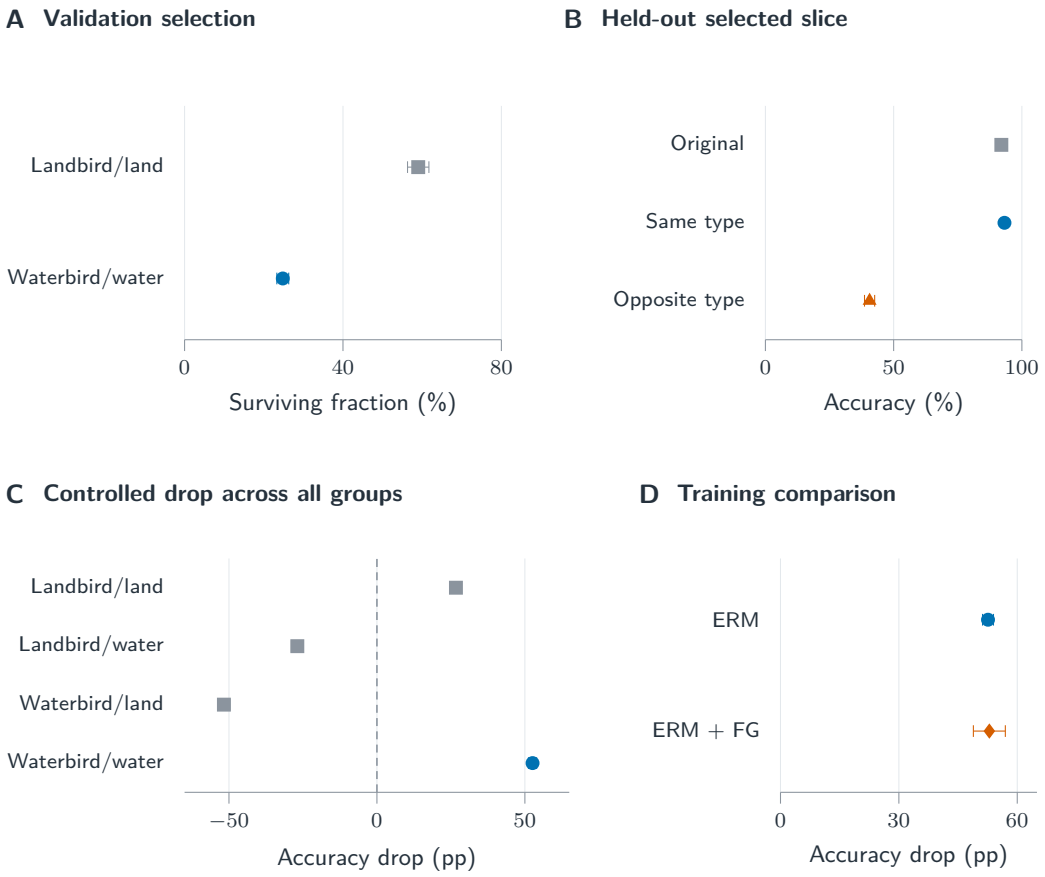


Figure D.1: **Validation selection and prospective background intervention.** (A) Uncentered feature–Schur fractions for the two eligible slices, with selected waterbird/water in blue. Eligibility requires accuracy at least the overall validation accuracy and NLL at most the median across slices; eligibility and selection are evaluated separately within each seed. (B) Held-out accuracy on the frozen selected slice: original images, same-type background control, and opposite-type replacement. (C) Controlled accuracy drops (same-type minus opposite-type) across all four groups; negative values indicate improvement under the opposite-type replacement. (D) The selected slice's controlled drop for ERM and validation-selected ERM+FG. Each model and slice is frozen before test interventions; the extracted bird foreground and labels are preserved. Points and bars show means and reported nominal 95% normal-approximation intervals (1.96 standard errors) across five seeds, 22–26. The selected-slice ERM drop is 52.6 ± 1.4 percentage points. The ERM+FG comparison is not accuracy-matched in two of five seeds.

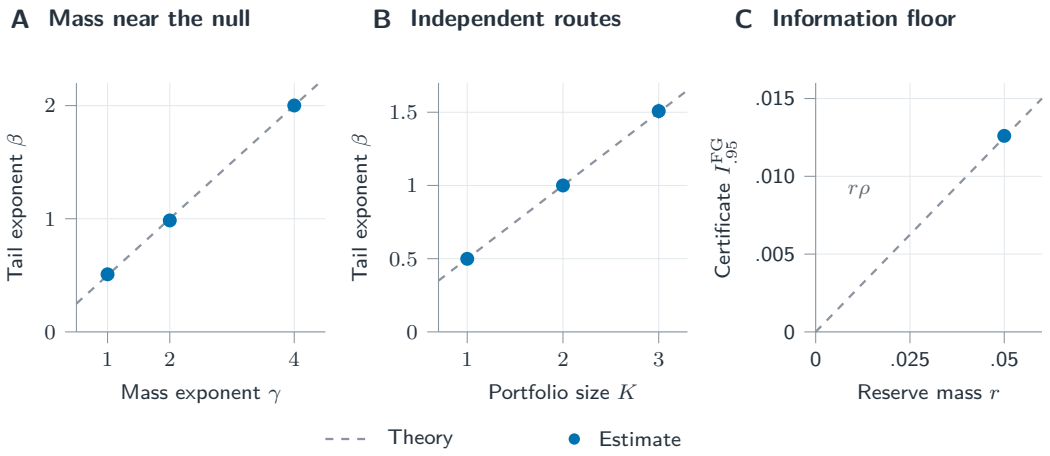


Figure D.2: **Structural checks for inverse-information tails.** (A) The reported single-seed exponents for mass parameters $\gamma = 1, 2, 4$ follow $\beta = \gamma/2$. (B) Independent portfolios follow $\beta = K/2$; points and intervals are the reported five-seed means and nominal 95% normal-approximation intervals (± 1.96 standard errors). (C) A reserve with mass $r = 0.05$ and anchor information $\rho = 0.25$ has reported certificate 0.01260 ± 0.000001, above its theoretical floor $r\rho = 0.0125$. The dashed curve is the analytic floor; only one reserve setting was evaluated. Intervals smaller than the markers remain plotted. All certificates use $\alpha = 0.95$.

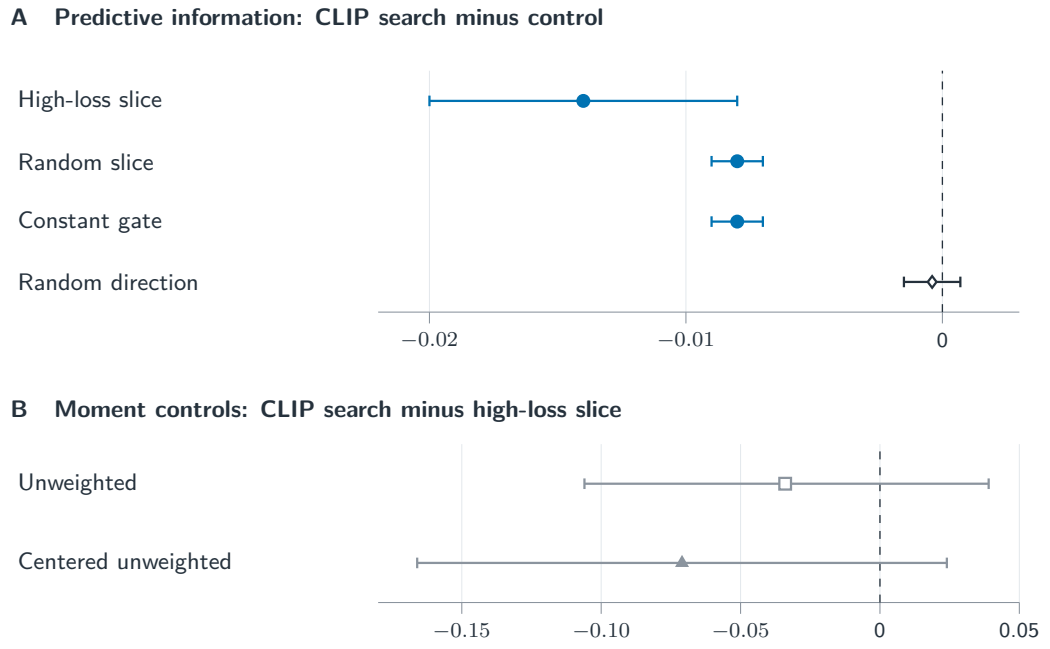


Figure D.3: **Paired discovery contrasts and moment controls.** Each point is a within-seed difference averaged across five paired seeds, with a two-sided 95% Student-t interval. Negative values indicate lower projected information for the CLIP-direction Fisher search. Predictive information is lower than for high-loss and random selectors with the same target within-class mass and the constant feasible gate (A). The random-direction comparison and both unweighted controls (B) include zero. Panel B reevaluates the predictive-selected gates under alternative matrices and uses a separate scale. These controls leave the incremental contribution of nuisance semantics unresolved.

E Information-Changing Interventions

This appendix develops the interventions in Section 6. We establish an information floor for anchor allocation, specify regularized training and model-selection protocols, derive tail-transverse influence, and report acquisition outcomes. The training, repair, and acquisition comparisons assess projected information together with predictive performance under their declared matrix and coordinate conventions.

E.1 Information Reserves under Observed-Source Allocation

We specialize the Fisher-reserve principle of Mansouri and Arai [2026] to an observed-source mixture of joint information matrices. Fix the predictor, task and nuisance coordinates, and their normalization. Let J_e denote the joint information in environment e, and let J_0 denote the anchor information in the same coordinates. Assume both matrices are positive semidefinite with positive-definite nuisance blocks.

Consider an observed-source design that samples the base experiment with probability $1-r$ and the anchor with probability $r \in (0,1]$, with allocation probabilities independent of the task and nuisance parameters. The source identity is part of each observation. Its likelihood score is the score of the observed component, so the per-observation information is

$$J_e^{\mathrm{res}} = (1-r)J_e + rJ_0. \tag{E.1}$$

The same affine relation holds for an allocation of declared operational matrix contributions with a fixed matrix map (Appendix B.1). This information-accumulation relation specifies the experiment to which the reserve guarantee applies.

Write $F_e^0 = S_0(J_e)$ and $F_0^0 = S_0(J_0)$ using the unregularized Schur operator from Appendix B.2. Suppose the anchor satisfies $F_0^0 \succeq \rho_0 I$ for some $\rho_0 > 0$. Then, for every environment e,

$$\boxed{F_e^{\mathrm{res}} := S_0(J_e^{\mathrm{res}}) \succeq (1-r)F_e^0 + rF_0^0 \succeq r\rho_0 I.} \tag{E.2}$$

Proof. For a block matrix $J = \left(\begin{smallmatrix} A & B \\ B^\top & C \end{smallmatrix}\right)$ with $C \succ 0$, define

$$q_J(u,v) = \begin{pmatrix} u \\ -v \end{pmatrix}^\top J \begin{pmatrix} u \\ -v \end{pmatrix}. \tag{E.3}$$

Completing the square gives

$$\begin{aligned} q_J(u,v) &= u^\top S_0(J)u \\ &\quad + (v - C^{-1}B^\top u)^\top C(v - C^{-1}B^\top u), \end{aligned} \tag{E.4}$$

and hence $u^\top S_0(J)u = \min_{v\in\mathbb{R}^{d_N}} q_J(u,v)$. For every fixed u, linearity of q_J in J yields

$$\begin{aligned} &u^\top S_0((1-r)J_e + rJ_0)u \\ &\quad = \min_v \{(1-r)q_{J_e}(u,v) + rq_{J_0}(u,v)\} \\ &\quad \geq (1-r)\min_v q_{J_e}(u,v) + r\min_v q_{J_0}(u,v) \\ &\quad = u^\top\big((1-r)F_e^0 + rF_0^0\big)u. \end{aligned} \tag{E.5}$$

This proves Schur-complement concavity in Loewner order. The information floor follows from $F_e^0 \succeq 0$ and the anchor bound. □

For any declared environment distribution Π, $0 < \alpha < 1$, and stabilization $\varepsilon \geq 0$, the pointwise floor implies

$$L_e^{\mathrm{res},(\varepsilon)} = \frac{1}{\lambda_{\min}(F_e^{\mathrm{res}}) + \varepsilon} \leq \frac{1}{r\rho_0 + \varepsilon}. \tag{E.6}$$

Consequently, monotonicity of CVaR (Appendix A.4) gives

$$\begin{aligned} I_\alpha^{\mathrm{FG,res}}(\varepsilon) &:= \left[\mathrm{CVaR}_{\alpha,e\sim\Pi}\big(L_e^{\mathrm{res},(\varepsilon)}\big)\right]^{-1}, \\ &\boxed{I_\alpha^{\mathrm{FG,res}}(\varepsilon) \geq r\rho_0 + \varepsilon \geq r\rho_0.} \end{aligned} \tag{E.7}$$

The final expression is the coarser bound reported in Section 6. For n base observations and m anchor observations, the normalized allocation is $r = m/(n+m)$. The guarantee is evaluated at the fixed predictor and coordinate system; a subsequent training step is followed by a new audit of the resulting predictor.

An estimated anchor can be incorporated through a positive lower confidence bound $\underline{\rho}_0 \le \lambda_{\min}(F_0^0)$ from Appendix B.3, under that appendix's estimation and conditioning assumptions. On its confidence event, the reserve bounds hold with $\rho_0 = \underline{\rho}_0$ simultaneously across the base environments.

In the synthetic construction, anchor amplitude $\delta_0 = 0.5$ supplies projected information $\rho_0 = \delta_0^2 = 0.25$. The allocation $r = 0.05$ gives $r\rho_0 = 0.0125$. The reported five-seed certificate is 0.01260 ± 0.000001 at $\alpha = 0.95$ and $\varepsilon = 10^{-12}$, where the uncertainty is the nominal 95% normal-approximation interval half-width. Appendix D.3 specifies this evaluated reserve setting.

E.2 Regularized Training

Regularized training combines a predictive objective with the information criterion in Sections 3–4. For training group g, let $\widehat{J}_g(w)$ be the declared operational matrix at predictor parameters w, and define

$$\widehat{F}_g^{\mathrm{rid}}(w) = S_{\mathrm{rid}}(\widehat{J}_g(w)), \qquad \widehat{L}_g(w) = \left[\lambda_{\min}(\widehat{F}_g^{\mathrm{rid}}(w)) + \varepsilon\right]^{-1}. \tag{E.8}$$

For fixed group weights $\pi_g \ge 0$ with $\sum_g \pi_g = 1$, the objective is

$$\begin{aligned} \min_{w,\tau} \quad & \widehat{R}_{\mathrm{base}}(w) \\ & + \lambda_{\mathrm{FG}} \left[\tau + \frac{1}{1-\alpha} \sum_{g=1}^{G} \pi_g \big(\widehat{L}_g(w) - \tau\big)_+\right]. \end{aligned} \tag{E.9}$$

Here $\lambda_{\mathrm{FG}} \ge 0$, and $\widehat{R}_{\mathrm{base}}$ is empirical task risk or the Group DRO objective. Uniform weights $\pi_g = 1/G$ give Equation 11. The information penalty increases the cost of weak projected directions, while the base objective specifies predictive performance. With four equally weighted Waterbirds groups and $\alpha = 0.95$, this CVaR is the maximum group inverse-information loss. The implemented frozen-feature training tail level is 0.75, which also gives that maximum for four equally weighted groups. Certificates are evaluated at $\alpha = 0.95$.

The operational matrix, coordinate normalization, ridge, and inverse-information stabilization determine the penalty's scale. In this known-group experiment, the operational matrix is the unweighted second moment of the standardized current task logit and fixed background-probe score, rather than the predictive-$p(1-p)$ matrix used in hidden-slice discovery. The nuisance ridge is $\lambda_{\mathrm{rid}} = 10^{-3}$ and the reciprocal stabilization is $\varepsilon = 10^{-8}$.

Frozen-feature protocol and coefficient selection. The experiment uses cached ImageNet MobileNetV3-small features and the Waterbirds diagnostic's background-probe nuisance coordinate. Feature and probe-score standardization use training moments. The current task logits are standardized during training by their full-training mean and population standard deviation plus 10^{-6}. A linear head is trained by full-batch AdamW for 1,200 updates, with learning rate 0.01, weight decay 10^{-4}, seeds 17–21, weights initialized as $N(0, 0.01^2)$, and zero intercept. Evaluation normalizes task logits by the resulting training-logit moments. The training implementation clamps each projected information value below at ε before computing $1/(F+\varepsilon)$; this extra training guard is distinct from the evaluation matrix definition.

The Group DRO baseline minimizes the maximum of the four group mean binary cross-entropies. Loss-CVaR training averages the largest $\lceil 0.1n \rceil$ per-example losses; its reported evaluation loss-CVaR uses level 0.95. JTT first fits ERM and then retrains with weight 20 on its training mistakes, normalized by total example weight. The reported runs use an Intel i9-11900F workstation with 32 GB RAM. Official training, validation, and test splits retain their fitting, selection, and final-evaluation roles.

For Fisher-Glass and Group DRO + FG, the reported coefficient grid is

$$\Lambda_{\mathrm{FG}} = \{0.01, 0.03, 0.1, 0.3\}. \tag{E.10}$$

For each seed and family, let $\widehat{a}_{\mathrm{worst,val}}(\lambda)$ denote validation worst-group accuracy. Define

$$\begin{aligned} a_\star &= \max_{\lambda \in \Lambda_{\mathrm{FG}}} \widehat{a}_{\mathrm{worst,val}}(\lambda), \\ \Lambda_{\mathrm{adm}} &= \{\lambda \in \Lambda_{\mathrm{FG}} : \widehat{a}_{\mathrm{worst,val}}(\lambda) \geq a_\star - 0.02\}, \\ \widehat{\lambda}_{\mathrm{FG}} &\in \arg\max_{\lambda \in \Lambda_{\mathrm{adm}}} \widehat{I}^{\mathrm{rid}}_{0.95,\mathrm{val}}(\lambda). \end{aligned} \tag{E.11}$$

The tolerance is two percentage points; certificate ties prefer higher validation worst-group accuracy. This rule uses validation group annotations. In the original reported cohort, it selects 0.03 for standalone Fisher-Glass in all five seeds and coefficients in $\{0.1, 0.3\}$ for Group DRO + FG.

Frozen-feature results. Tables E.1 and E.2 give the complete six-method comparison. Entries are means with nominal 95% normal-approximation intervals (± 1.96 standard errors) across five training seeds; information certificates use $\alpha = 0.95$.

Table E.1: Frozen-feature Waterbirds: predictive performance in the original training cohort.

Method	Average acc. (%)	Worst-group acc. (%)	Loss-CVaR$_{0.95}$
ERM	77.2 ± 0.1	55.7 ± 0.3	1.969 ± 0.007
Group DRO	80.9 ± 0.3	67.3 ± 0.3	1.143 ± 0.021
JTT	77.4 ± 0.1	54.1 ± 0.2	2.167 ± 0.004
Loss-CVaR	78.2 ± 0.1	52.1 ± 0.3	2.427 ± 0.041
Fisher-Glass	80.4 ± 0.2	56.3 ± 0.3	1.866 ± 0.025
Group DRO + FG	80.6 ± 0.9	69.4 ± 1.3	0.855 ± 0.082

Table E.2: Frozen-feature Waterbirds: ridge information in the original training cohort.

Method	Ridge certificate	Inverse-information p95
ERM	0.198 ± 0.001	4.86 ± 0.02
Group DRO	0.233 ± 0.003	4.10 ± 0.06
JTT	0.234 ± 0.001	4.13 ± 0.01
Loss-CVaR	0.233 ± 0.002	4.17 ± 0.03
Fisher-Glass	0.399 ± 0.006	2.46 ± 0.04
Group DRO + FG	0.614 ± 0.059	1.63 ± 0.17

Fisher-Glass increases the mean certificate from 0.198 to 0.399 relative to ERM. Adding the penalty to Group DRO increases the certificate from 0.233 to 0.614 and mean worst-group accuracy from 67.3% to 69.4%. JTT and loss-CVaR supply comparisons with error-based and predictive-loss objectives. Predictive performance and projected information induce distinct rankings.

These tables preserve the original reported training cohort. Appendix B provides ridge-to-unregularized comparisons and lower audits for fresh replications with retained group-level blocks; those runs do not retroactively certify the historical checkpoints. The implementation is `experiments/waterbirds_fisher_glass_training.py`.

E.3 Extended Training and Selection Results

We examine four settings with distinct training budgets, information targets, and selection rules: an end-to-end penalty sweep, the prospective background intervention, constrained repair of discovered slices, and coefficient selection without group annotations. Each comparison records information and predictive performance under its own evaluation protocol.

End-to-end penalty frontier. ImageNet-initialized MobileNetV3-small Howard et al. [2019] is fine-tuned on Waterbirds for 10 epochs with training batch size 128, evaluation batch size 512, and seeds 17–21. The reported configuration uses CUDA bf16 automatic mixed precision, TF32, channels-last tensors, and eight data-loader workers on an NVIDIA H100 MIG slice with approximately 47 GB memory. AdamW uses learning rate 10^{-4} and weight decay 10^{-4}; the final epoch supplies the evaluated model. The nuisance coordinate is the binary background annotation standardized by

training moments. The minibatch task logit is standardized by its mean and population standard deviation plus 10^{-6}. Groups with fewer than two batch observations are omitted from the information penalty, and the remaining groups' upper-tail loss is computed at $\alpha = 0.95$. The nuisance ridge is 10^{-3}; projected information is clamped below at 10^{-8} before another 10^{-8} is added in the reciprocal.

The six Group DRO + FG coefficients are

$$\lambda_{\mathrm{FG}} \in \{5\times10^{-5}, 10^{-4}, 3\times10^{-4}, 10^{-3}, 3\times10^{-3}, 10^{-2}\}. \tag{E.12}$$

Tables E.3 and E.4 report the full frontier. FG entries are means $\pm$ standard deviations over five seeds. The Group DRO reference reproduces the reported comparison at $\lambda_{\mathrm{FG}} = 10^{-4}$; each sweep file reran its own Group DRO baseline, so this single reference is not a common paired baseline for every coefficient.

Table E.3: End-to-end Waterbirds: predictive performance over the penalty frontier.

Objective / λ_{FG}	Average acc. (%)	Worst-group acc. (%)	Loss-CVaR$_{0.95}$
Group DRO reference	85.4	53.2	0.925
5×10^{-5}	79.7 ± 6.2	56.6 ± 5.8	0.733 ± 0.061
10^{-4}	79.5 ± 10.2	57.7 ± 8.3	0.733 ± 0.110
3×10^{-4}	69.9 ± 9.7	58.8 ± 8.7	0.653 ± 0.064
10^{-3}	62.7 ± 7.9	49.9 ± 1.8	0.710 ± 0.012
3×10^{-3}	57.7 ± 4.5	48.6 ± 6.6	0.717 ± 0.036
10^{-2}	47.9 ± 16.5	22.7 ± 16.7	1.031 ± 0.416

Table E.4: End-to-end Waterbirds: ridge information over the penalty frontier.

Objective / λ_{FG}	Ridge certificate	Inverse-information p95
Group DRO reference	0.259	3.68
5×10^{-5}	0.393 ± 0.059	2.49 ± 0.34
10^{-4}	0.502 ± 0.166	2.10 ± 0.58
3×10^{-4}	0.614 ± 0.189	1.71 ± 0.46
10^{-3}	0.786 ± 0.124	1.29 ± 0.20
3×10^{-3}	0.817 ± 0.068	1.22 ± 0.10
10^{-2}	0.809 ± 0.076	1.23 ± 0.12

At $\lambda_{\mathrm{FG}} = 10^{-4}$, the mean certificate is 0.502 and inverse-information p95 is 2.10, compared with 0.259 and 3.68 for its reference; mean average and worst-group accuracies are 79.5% and 57.7%. Larger coefficients increase the certificate over much of the grid while average accuracy decreases. The largest coefficient yields 22.7% mean worst-group accuracy. Figure E.1 presents this frontier, making coefficient choice an explicit decision about information and performance.

The runs use `experiments/end_to_end_deep_suite.py`, launched by `experiments/run_full_end_to_end_suite.sh`. The reported batch sizes override that launcher's defaults of 32 and 256. Historical sweep summaries are retained in `reports/fisher_glass_results_report.md`; Appendix B's fresh audit runs form a separate cohort.

Training in the prospective intervention study. The prospective protocol in Appendix D.2 uses ImageNet-initialized MobileNetV3-small models fully fine-tuned for 50 epochs, batch size 32, AdamW learning rate 10^{-4} and weight decay 10^{-4}, bf16 on an H100, and seeds 22–26. The FG arm evaluates coefficients $\{0.003, 0.01, 0.03\}$. During training, a linear background probe is fitted by AdamW at learning rate 10^{-3} on detached current features; its continuous score is detached before entering the penalty. Its optimizer uses the library's default weight decay 0.01. The penalty is disabled for the first epoch. Batch-standardized task and nuisance scores use a 10^{-6} standard-deviation guard. A group contributes only with at least six examples and nuisance standard deviation at least 0.05. The training Schur value is floored at 0.05 and its reciprocal is taken directly; this optimization floor is separate from the audit's $\varepsilon = 10^{-8}$. Training uses tail level 0.75 over available groups, with ridge 10^{-3}.

The final trained checkpoint is audited on validation data. For each seed, coefficient selection maximizes the validation certificate among candidates with overall validation accuracy at least the ERM value minus 0.01. If that set is empty, the implementation maximizes the certificate over all candidates and records a fallback. The selected coefficients for seeds 22–26 are $[0.01, 0.03, 0.01, 0.003, 0.01]$; seeds 22 and 24 fall back. The audit uses $\alpha = 0.95$, ridge 10^{-3}, and $\varepsilon = 10^{-8}$. The checkpoint, calibration, independent audit probe, selected slice, and audit constants are fixed before intervention scoring.

In the current rerun, FG training increases the model-level validation certificate from 0.289 to 0.330. Its selected slice has a controlled same-type-minus-opposite-type accuracy drop of 52.9 percentage points, compared with 52.6 for ERM. Among predictions correct on original images, 57.7% flip under the opposite-type intervention in the FG arm, compared with 56.4% for ERM. Accuracy matching fails in two seeds. The information increase therefore accompanies persistent sensitivity to the audited background change. Appendix D.2 gives the preserved-foreground construction, artifact control, and selection history; these values use `reports/prospective_nuisance_shift_report_v2_rerun.md`, whose ERM opposite-type accuracy is 40.6%.

Constrained repair of discovered slices. The hidden-slice study evaluates predictive-loss, Fisher-information, and combined pseudo-group repair. Candidates must keep overall, balanced, and each task-class validation accuracy within 0.02 of the base model. These screens use task labels, while background and group annotations remain sealed. Every arm includes an exact unchanged-model candidate. Loss coefficients are $\{0.1, 0.3, 1\}$, Fisher coefficients $\{0.01, 0.03, 0.1\}$, and combined repair uses their Cartesian product.

Each active candidate starts from the same ERM head and uses a fresh AdamW optimizer, learning rate 0.01, weight decay 10^{-4}, gradient-norm clipping at 10, and at most 1,200 additional full-batch steps. Binary NLL at temperature one on the held-out training-calibration partition is checked every 25 steps; improvement must exceed 10^{-5}, patience is 12 checks, and the best checkpoint is restored with step zero eligible. The ERM calibration temperature remains fixed for Fisher training, validation selection, rediscovery, and test auditing. The matched ERM continuation has the same warm start, extra-step budget, optimizer, and checkpoint rule with both penalties zero.

The Fisher term is dimensionless. Writing F_{init} for the predictive projected information on the frozen training slice at the warm start, the implemented penalty is

$$P_{\mathrm{FG}}(w) = \frac{\max\{F_{\mathrm{init}}, \varepsilon\}}{F(w) + \varepsilon}, \qquad \varepsilon = 10^{-8}. \tag{E.13}$$

It is approximately one at the warm start when $F_{\mathrm{init}} \gg \varepsilon$; stabilization prevents exact equality. The scalar predictive matrix retains shrinkage 0.1 and ridge 10^{-3}, with negative numerical projected values clipped to zero.

Within the accuracy envelope, loss repair minimizes fixed-temperature loss on the frozen original slice, and Fisher repair maximizes its predictive projected information. Combined repair retains the loss–information Pareto frontier, then maximizes the smaller relative improvement over the base slice; ties use the mean relative improvement, smaller normalized coefficient sum, and earlier candidate index. Single-criterion ties similarly favor smaller penalties and earlier indices. After selection, each head receives fresh slice rediscovery with matched gate-initialization and fold seeds. Thus the slice evaluated below can change after training.

Table E.5: Constrained hidden-slice repair: predictive performance and selected action.

Training arm	Average acc. (%)	Worst-group acc. (%)	Unchanged selections
Matched ERM continuation	76.3 ± 0.7	57.3 ± 1.3	—
Loss repair	76.3 ± 0.6	56.9 ± 1.3	$3/5$
Fisher repair	76.3 ± 0.7	57.3 ± 1.3	$5/5$
Combined repair	76.4 ± 0.9	56.7 ± 1.6	$4/5$
Oracle Group DRO	80.0 ± 0.3	64.8 ± 0.9	—

Accuracy, NLL, and information entries are means $\pm 95\%$ Student-t interval half-widths across five paired seeds; composition entries are mean percentages. The oracle uses group annotations during

Table E.6: Constrained repair: the CLIP-defined slice rediscovered after each arm. Information and NLL are evaluated on that arm's rediscovered slice.

Training arm	Slice NLL	Predictive projected info.	Landbird/land mass
Matched ERM continuation	0.031 ± 0.017	0.005 ± 0.001	97.0%
Loss repair	0.023 ± 0.017	0.005 ± 0.002	97.8%
Fisher repair	0.031 ± 0.017	0.005 ± 0.001	97.0%
Combined repair	0.031 ± 0.017	0.005 ± 0.001	97.2%
Oracle Group DRO	0.033 ± 0.018	0.005 ± 0.002	95.9%

training. Fisher repair selects the unchanged model in all five seeds; loss-only and combined repair do so in three and four. Paired worst-group accuracy changes relative to matched continuation are -0.4, 0.0, and -0.6 percentage points for loss, Fisher, and combined repair, with intervals containing zero. The oracle reaches 64.8% worst-group accuracy while rediscovered projected information remains 0.005. These results record both the action and the population on which information is assessed. The implementation is `experiments/waterbirds_hidden_slice`.

The matching retained cohort is `experiments/waterbirds_hidden_slice_v2_corrected_results.tar.gz`, rooted at `waterbirds_hidden_slice_v2/`, with `main_table.csv` and `waterbirds_hidden_slice_report.md` as described in Appendix D.4.

Post hoc coefficient selection without group labels. This sensitivity analysis uses $\Lambda_{\text{hidden}} = \{0, 0.01, 0.03, 0.1, 0.3\}$ with the dimensionless normalization in Equation E.13. Fixed 0.03 is a reference within this protocol and does not reproduce the known-group objective's numerical scale. Training and selection use task labels and the fixed CLIP water-versus-land nuisance coordinate. Training data are partitioned into 70% fitting, 15% calibration, and 15% gate discovery. A frozen train-only slice supplies the candidate penalty. Every candidate, including $\lambda = 0$, receives the same warm start, head-training seed, AdamW settings, maximum 1,200-step budget, and calibration-NLL checkpoint rule described above. One base ERM temperature is frozen across candidates. Official validation observations are used only after candidate heads are fixed. Background and group metadata remain sealed through blind test evaluation.

Let n_v denote validation size and $\ell_{i,\lambda}$ the fixed-temperature per-example validation NLL. Define

$$\widehat{R}_\lambda = \frac{1}{n_v} \sum_i \ell_{i,\lambda}, \qquad s_\lambda = \sqrt{\frac{\sum_i (\ell_{i,\lambda} - \widehat{R}_\lambda)^2}{n_v(n_v - 1)}}. \tag{E.14}$$

The risk minimizer λ_R minimizes $(\widehat{R}_\lambda, \lambda)$ lexicographically. The rule uses the standard error of this minimizer for every candidate:

$$\Lambda_{\text{risk}} = \{\lambda : \widehat{R}_\lambda \le \widehat{R}_{\lambda_R} + s_{\lambda_R}\}. \tag{E.15}$$

The numerical eligibility check allows 10^{-15} slack. Candidate-specific hidden slices are rediscovered in five task-stratified validation folds. Let $V_{k,\lambda}$ be the held-out inverse predictive information in fold k, and let $s_{V,\lambda}$ be their sample standard deviation. With five coefficients and nominal confidence 0.95, the adjusted score is

$$\begin{aligned} U_\lambda^{\text{adj}} &= \overline{V}_\lambda + t_{4,\,1-0.05/5} \frac{s_{V,\lambda}}{\sqrt{5}}, \\ \underline{I}_{\text{hidden}}^{\text{adj}}(\lambda) &= (U_\lambda^{\text{adj}})^{-1}, \\ \widehat{\lambda}_{\text{hidden}} &= \min \arg \max_{\lambda \in \Lambda_{\text{risk}}} \underline{I}_{\text{hidden}}^{\text{adj}}(\lambda). \end{aligned} \tag{E.16}$$

Here $t_{4,0.99} = 3.74695$. The superscript identifies Bonferroni adjustment over the coefficient grid. Because gate-training folds overlap, this Student-t-shaped quantity is a validation ranking score, without a formal independent-fold coverage claim. Overall, balanced, and per-class two-percentage-point accuracy screens and the four positive-coefficient paired inverse-information comparisons are diagnostics only and do not alter the choice. The analysis was designed after earlier group/test outcomes had been inspected on this benchmark.

Table E.7: Post hoc group-free coefficient selection: test predictive performance. Brackets are two-sided 95% Student-t intervals over five paired seeds.

Policy	Average accuracy (%)	Worst-group accuracy (%)
Frozen ERM reference	75.5 [74.3, 76.6]	55.7 [53.3, 58.2]
Matched continuation, $\lambda = 0$	75.5 [74.3, 76.6]	55.7 [53.3, 58.2]
Fixed $\lambda = 0.03$	76.6 [76.1, 77.2]	51.5 [47.1, 55.8]
Validation-risk only	77.5 [76.5, 78.6]	43.1 [33.1, 53.1]
Group-free selector	77.5 [76.5, 78.6]	43.1 [33.1, 53.1]
Group-supervised comparator	77.7 [76.6, 78.8]	44.2 [34.2, 54.3]

Table E.8: Post hoc coefficient choices and hidden information on the corresponding rediscovered test slices. Intervals describe across-seed variation on the fixed benchmark.

Policy	Coefficients: seed counts	Hidden information
Frozen ERM reference	—	0.007 [0.005, 0.009]
Matched continuation, $\lambda = 0$	$0:5$	0.007 [0.005, 0.009]
Fixed $\lambda = 0.03$	$0.03:5$	0.007 [0.004, 0.010]
Validation-risk only	$0:1,\ 0.3:4$	0.012 [0.006, 0.019]
Group-free selector	$0:1,\ 0.3:4$	0.012 [0.006, 0.019]
Group-supervised comparator	$0:1,\ 0.1:1,\ 0.3:3$	0.011 [0.005, 0.018]

The group-supervised comparator accesses validation group annotations after group-free decisions and blind test artifacts are fixed; it is a diagnostic comparator, not an upper bound. The selector chooses 0.3 in four seeds and zero in one, exactly matching plain validation-NLL minimization in every seed. Relative to matched continuation, the paired mean changes are $+2.1$ percentage points in average accuracy (95% interval $[+0.2, +4.0]$), $+0.006$ hidden information ($[+0.001, +0.010]$), and -12.6 points in worst-group accuracy ($[-21.6, -3.7]$). These contrasts describe the complete tested protocol; its information criterion does not change the risk-only choices. The matched zero-coefficient continuation restores step zero in all five seeds and consequently agrees with the frozen ERM reference. The study makes the validation objective central to information-directed training. Its implementation is `experiments/waterbirds_group_free_lambda`.

Both tables use `experiments/waterbirds_group_free_lambda_v2_results_full.tar.gz`, rooted at `waterbirds_group_free_lambda_v2/`, specifically `main_table.csv` and `report.md`.

E.4 Tail-Transverse Influence

An acquisition score evaluates how a candidate changes the current information audit. Fix the predictor, coordinates, calibration, slice rules, and finite audit weights $\pi_g \geq 0$ summing to one. For action a, specify an admissible positive-semidefinite matrix path

$$J_g(t, a) = J_g + tH_{g,a} + o(t), \qquad t \downarrow 0, \tag{E.17}$$

where the remainder is in operator norm and $H_{g,a}$ is symmetric. The path incorporates allocation, normalization, and any retained pooling in the declared matrix target. Audit weights remain fixed along this path.

Schur differential and the lifted direction. Write

$$J_g = \begin{pmatrix} A_g & B_g \\ B_g^\top & C_g \end{pmatrix}, \quad C_g \succ 0, \qquad H = \begin{pmatrix} H_{TT} & H_{TN} \\ H_{TN}^\top & H_{NN} \end{pmatrix}. \tag{E.18}$$

Differentiating the inverse in the Schur operator of Appendix B.2 gives

$$\begin{aligned} DS_0(J_g)[H] = H_{TT} - H_{TN}C_g^{-1}B_g^\top - B_gC_g^{-1}H_{TN}^\top \\ + B_gC_g^{-1}H_{NN}C_g^{-1}B_g^\top. \end{aligned} \tag{E.19}$$

Suppose $\lambda_g = \lambda_{\min}(S_0(J_g))$ is simple, with unit eigenvector u_g. Define the lifted efficient direction

$$p_g = \begin{pmatrix} u_g \\ -C_g^{-1} B_g^\top u_g \end{pmatrix}. \tag{E.20}$$

Differentiating the eigenvalue equation and using $u_g^\top u_g = 1$ gives $D\lambda_g[H] = u_g^\top DS_0(J_g)[H]u_g$. Expanding $p_g^\top H p_g$ yields the four terms in this quadratic form. Therefore,

$$\boxed{h_{g,a} := \frac{d}{dt}\lambda_{\min}(S_0(J_g(t,a)))\Big|_{t=0+} = p_g^\top H_{g,a} p_g.} \tag{E.21}$$

For an additive rank-one contribution $H_{g,a} = s_a s_a^\top$,

$$\begin{aligned} h_{g,a} &= (p_g^\top s_a)^2, \\ p_g^\top s_a &= u_g^\top (s_{T,a} - B_g C_g^{-1} s_{N,a}). \end{aligned} \tag{E.22}$$

This measures the candidate score remaining along the weak task direction after nuisance residualization.

Allocation and normalization. For normalized augmentation by candidate matrix $M_{g,a}$,

$$J_g(t,a) = \frac{J_g + tM_{g,a}}{1+t}, \qquad H_{g,a} = M_{g,a} - J_g. \tag{E.23}$$

The variational identity (E.4) gives $p_g^\top J_g p_g = \lambda_g$, so

$$h_{g,a} = p_g^\top M_{g,a} p_g - \lambda_g. \tag{E.24}$$

A replacement has perturbation equal to its new contribution minus its removed contribution, with the declared allocation rate included. These expressions place candidate gains in the same information units as the audited matrices.

Fixed-ridge derivative. For S_{rid} from Appendix B.2, hold λ_{rid} fixed and set

$$Q_g = (C_g + \lambda_{\mathrm{rid}} I)^{-1}, \qquad p_{g,\mathrm{rid}} = \begin{pmatrix} u_{g,\mathrm{rid}} \\ -Q_g B_g^\top u_{g,\mathrm{rid}} \end{pmatrix}, \tag{E.25}$$

where $u_{g,\mathrm{rid}}$ is a unit eigenvector associated with a simple smallest eigenvalue $\lambda_{g,\mathrm{rid}}$ of $S_{\mathrm{rid}}(J_g)$. Since $DQ_g[H_{NN}] = -Q_g H_{NN} Q_g$, the Schur differential takes the same form with C_g^{-1} replaced by Q_g. Thus

$$D\lambda_{g,\mathrm{rid}}[H] = p_{g,\mathrm{rid}}^\top H p_{g,\mathrm{rid}}. \tag{E.26}$$

For normalized augmentation, the identity $Q_g C_g Q_g = Q_g - \lambda_{\mathrm{rid}} Q_g^2$ yields

$$p_{g,\mathrm{rid}}^\top J_g p_{g,\mathrm{rid}} = \lambda_{g,\mathrm{rid}} - \lambda_{\mathrm{rid}} \|Q_g B_g^\top u_{g,\mathrm{rid}}\|_2^2. \tag{E.27}$$

Consequently,

$$\begin{aligned} D\lambda_{g,\mathrm{rid}}[M_{g,a} - J_g] = p_{g,\mathrm{rid}}^\top M_{g,a} p_{g,\mathrm{rid}} - \lambda_{g,\mathrm{rid}} \\ + \lambda_{\mathrm{rid}} \|Q_g B_g^\top u_{g,\mathrm{rid}}\|_2^2. \end{aligned} \tag{E.28}$$

The ridge gain therefore uses its own eigenvalue, eigenvector, lifted direction, and allocation derivative consistently.

CVaR sensitivity. Choose either the exact or fixed-ridge matrix target consistently, and write its eigenvalues and derivatives as λ_g and $h_{g,a}$. Fix $\varepsilon > 0$ and $0 < \alpha < 1$. With $L_g = (\lambda_g + \varepsilon)^{-1}$, define the feasible tail weights

$$\mathcal{W}_{\alpha,\pi} = \left\{ \omega \geq 0 : \sum_g \omega_g = 1, \quad \omega_g \leq \frac{\pi_g}{1-\alpha} \right\}. \tag{E.29}$$

The finite-audit representation (A.26) gives

$$C_\alpha(L) := \mathrm{CVaR}_{\alpha,\pi}(L_g) = \max_{\omega \in \mathcal{W}_{\alpha,\pi}} \sum_g \omega_g L_g. \tag{E.30}$$

Let $\Omega^\star_\alpha(L)$ denote its maximizing weight vectors. If the maximizing vector ω_α is unique and the smallest eigenvalues are simple, differentiation yields

$$\boxed{G_\alpha(a) := -D_a C_\alpha = \sum_g \omega_{\alpha,g} \frac{h_{g,a}}{(\lambda_g + \varepsilon)^2}.} \tag{E.31}$$

Indeed, $D_a L_g = -h_{g,a}/(\lambda_g + \varepsilon)^2$, and the unique active linear functional supplies the derivative of the CVaR maximization. The normalized weights include the upper-tail mass factor. A positive gain decreases inverse-information CVaR and increases its reciprocal:

$$D_a I_{\alpha,\pi} = I^2_{\alpha,\pi} G_\alpha(a), \qquad I_{\alpha,\pi} = C_\alpha(L)^{-1}. \tag{E.32}$$

Ties and repeated eigenvalues. At a repeated minimum eigenvalue, the exact right derivative uses its eigenspace $\mathcal{E}_g$:

$$h_{g,a} = \min_{\substack{u \in \mathcal{E}_g \\ \|u\|_2 = 1}} p_g(u)^\top H_{g,a} p_g(u), \tag{E.33}$$

where the lift corresponds to the chosen exact or ridge target. This is the smallest eigenvalue of the Schur differential restricted to $\mathcal{E}_g$.

At a CVaR tie, the right derivative is obtained from the currently active linear functionals:

$$-D^+_a C_\alpha = \min_{\omega \in \Omega^\star_\alpha(L)} \sum_g \omega_g \frac{h_{g,a}}{(\lambda_g + \varepsilon)^2}. \tag{E.34}$$

This follows because the right derivative of the maximum in (E.30) is the maximum of its active directional loss derivatives; negation turns that maximum into a minimum. Equations (E.33) and (E.34) specify the directional gain at either source of nondifferentiability, including when both occur. A fixed maximizing tail-weight choice also defines an acquisition score; equality to the directional decrease follows under the unique-weight conditions above.

Tail-transverse influence. For a candidate intervention with assigned task label y_a, define

$$\boxed{\mathrm{TTI}(a) = G_\alpha(a)\, \ell_w(x^{\mathrm{cf}}_a, y_a),} \qquad \ell_w(x, y) = -\log p_w(y \mid x). \tag{E.35}$$

The information factor emphasizes weak task directions in the current audit. The loss factor emphasizes candidates that the current predictor handles poorly. Their product ranks candidates under the declared acquisition budget. At nonsmooth points, the directional gain in (E.34) supplies an exact right-sensitivity convention; an implementation using a fixed tail choice records that choice.

Scores are evaluated at the fixed predictor, and selected candidates then enter training. Held-out evaluation measures the realized information and predictive-performance changes after finite augmentation and retraining. The reported feature-space experiment uses a normalized residual-square surrogate $\widetilde{G}_\alpha(a)$ in its product score. Appendix E.5 specifies that implemented score and its distinction from the directional CVaR decrease derived here.

E.5 Feature-Space Acquisition Experiments

We evaluate information-guided acquisition through controlled feature-space interventions on Waterbirds. Each candidate translates a frozen representation along a difference between same-label background-group means. Selection uses a geometric residual score, current task loss, or their product; evaluation measures information and accuracy after head retraining. Task and background annotations define the interventions and groups.

Candidate construction and data roles. Let v be a frozen ImageNet MobileNetV3-small representation Howard et al. [2019], standardized coordinatewise using training means and standard deviations. For task label y and background annotation $b \in \{0, 1\}$, let $\bar{v}_{y,b}$ be the corresponding official-training-split mean. At strength γ_{cf}, construct

$$v^{\mathrm{cf}} = v + \gamma_{\mathrm{cf}}\big(\bar{v}_{y,1-b} - \bar{v}_{y,b}\big), \qquad y^{\mathrm{cf}} = y. \tag{E.36}$$

Candidates come from the official validation split. Their assigned background is $b^{\mathrm{cf}} = 1 - b$, so their resulting group is $(y, 1 - b)$, even at fractional γ_{cf}. The code falls back to background-only training means if a label–background mean is unavailable; the reported $\gamma_{\mathrm{cf}} = 1$ run uses label-conditional shifts for every candidate. Its mean and 95th-percentile displacement norms are 9.297 and 9.463. These translations preserve the supplied task label by construction and provide a frozen-feature proxy for nuisance intervention; they do not establish pixel-level or semantic label preservation.

A balanced logistic background probe is fitted on training features and held fixed. Its decision score is centered and scaled by its training-score mean and standard deviation, and recomputed on each translated feature. Selected features and fixed labels are concatenated with the original training set. The representation and background probe remain fixed; the task head is refitted from scratch. Test examples supply final evaluation only. Validation examples selected for acquisition enter training, so subsequent validation metrics are not an independent evaluation of the acquired model.

For acquisition fraction $b_{\mathrm{acq}} \in \{0.01, 0.02, 0.05\}$, the code uses

$$k = \min\{\max(1, \mathrm{round}(b_{\mathrm{acq}} n_{\mathrm{train}})), n_{\mathrm{cand}}\}. \tag{E.37}$$

The original training size is $4{,}795$ and the validation pool has $1{,}199$ examples. Thus the three prescribed counts are 48, 96, and 240, yielding augmented training sizes $4{,}843$, $4{,}891$, and $5{,}035$. All candidates are eligible for the selectors in the tables. These counts follow the implemented rule and split sizes; the retained summaries do not contain the per-run selected-index or group-count records.

Starting model and implemented geometric score. The refitted balanced logistic head has 78.2% average test accuracy, 52.5% worst-group accuracy, ridge certificate $I^{\mathrm{rid}}_{0.95} = 0.209$, and inverse-information 95th percentile 4.683. Its weakest-information group is landbird/land. This is the acquisition baseline; the predefined-group diagnostic uses its own fitted head and probe.

At the current head, let z be its logit centered and scaled using original-training logits, and let q be the standardized background-probe score. Standard deviations below 10^{-6} are replaced by 1. For each original validation group g, the operational matrix is the equally weighted within-group second moment of (z, q), without a predictive factor $p(1 - p)$. With $\lambda_{\mathrm{rid}} = 10^{-3}$ and $\varepsilon = 10^{-8}$, set

$$\begin{aligned} \beta_g &= \frac{J_{TN,g}}{J_{NN,g} + \lambda_{\mathrm{rid}}}, \\ \lambda_g &= J_{TT,g} - \frac{J^2_{TN,g}}{J_{NN,g} + \lambda_{\mathrm{rid}}}, \qquad L_g = (\lambda_g + \varepsilon)^{-1}. \end{aligned} \tag{E.38}$$

The current Schur routine floors numerically negative information at zero. The selector forms $T = \{g : L_g \geq \mathrm{quantile}_{0.95}(L_1, \ldots, L_4)\}$, using NumPy's interpolated quantile and including every threshold tie. Its actual candidate score is

$$\widetilde{G}_\alpha(a) = \frac{\displaystyle\sum_{g \in T} (\lambda_g + \varepsilon)^{-2} \big(z^{\mathrm{cf}}_a - \beta_g q^{\mathrm{cf}}_a\big)^2}{\max\left\{\displaystyle\sum_{g \in T} (\lambda_g + \varepsilon)^{-2}, 10^{-12}\right\}}. \tag{E.39}$$

For a unique weakest group, T contains that group alone and the inverse-square factor cancels. The code applies every retained tail projector to every candidate, independently of the candidate's assigned group. It includes no group-allocation indicator, group-size factor, subtraction of existing information, or replacement term. The residual square is nonnegative; the score has no additional negative-gain clipping.

Equation E.39 is a geometric surrogate for the directional gain in Appendix E.4. At fixed coordinates, its residual square is the ridge-Schur directional contribution of an unnormalized rank-one addition.

Actual acquisition appends examples to one assigned group, and within-group normalized augmentation instead changes a group moment toward candidate information, with allocation and sample-count scaling. Those terms are absent from the implemented ranking. Threshold-inclusive tail ties also do not implement the fractional or direction-dependent CVaR weights needed in the general derivative. Finally, refitting the head and recomputing logit normalization change the audited coordinates. The selected score therefore is not the exact derivative of the eventual post-training certificate.

The empirical selector ranks by $\widetilde{\mathrm{TTI}}_\alpha(a) = \widetilde{G}_\alpha(a)\ell_w(v_a^{\mathrm{cf}}, y_a)$; the tilde distinguishes this implemented product from the exact directional-gain product based on G_α. High-loss and pure-tail controls rank by ℓ_w and $\widetilde{G}_\alpha$, respectively. The rank product multiplies separately rank-normalized geometric scores and losses: ascending `argsort` positions receive evenly spaced values from 0 to 1; equal values are not assigned averaged ranks. The uncertainty product uses $\widetilde{G}_\alpha(a)p_a(1-p_a)$, with $p_a = \mathrm{sigmoid}(\mathrm{clip}(f_w(v_a^{\mathrm{cf}}), -60, 60))$. Task loss is binary cross-entropy, with probabilities floored at 10^{-12} inside logarithms. Top-k selection reverses NumPy's ascending `argsort`; no separate stable tie-breaking convention is imposed.

Budget comparison at $\gamma_{\mathrm{cf}} = 1$. Table E.9 retains all three supplied budgets and the selected-candidate summaries. Entries are means across seeds 17–21. Worst-group accuracy gains are percentage-point changes after acquisition and retraining; certificate gains are absolute changes in the ridge certificate. The geometric score and loss are measured before retraining on the selected candidates.

Table E.9: Acquisition at $\gamma_{\mathrm{cf}} = 1$. $\widetilde{\mathrm{TTI}}_\alpha$ denotes the implemented residual-score surrogate times task loss. Δa is worst-group accuracy gain in percentage points; ΔI is the absolute ridge-certificate gain. The retained report supplies means without uncertainty intervals.

Budget	Selector	Δa (pp)	ΔI	$\widetilde{G}_\alpha$	Loss
1%	$\widetilde{\mathrm{TTI}}_\alpha$	+4.52	+0.0317	1.046	8.159
	High loss	+3.27	+0.0603	0.471	11.913
	Pure tail	+0.62	+0.0045	2.946	0.176
2%	$\widetilde{\mathrm{TTI}}_\alpha$	+6.85	+0.0452	0.777	6.842
	High loss	+5.76	+0.0581	0.450	9.038
	Pure tail	+2.80	+0.0098	2.402	0.303
5%	$\widetilde{\mathrm{TTI}}_\alpha$	+7.94	+0.0606	0.557	4.245
	High loss	+7.01	+0.0630	0.403	4.794
	Pure tail	+3.58	+0.0340	1.691	1.038

The empirical $\widetilde{\mathrm{TTI}}_\alpha$ has larger reported mean worst-group accuracy gains than high-loss selection at all three budgets at $\gamma_{\mathrm{cf}} = 1$, while high loss produces larger certificate gains. At the 1% budget, pure-tail selection attains mean geometric score 2.946 but mean loss only 0.176. Multiplication by loss selects candidates with geometric score 1.046 and loss 8.159, producing a 4.52-point accuracy gain rather than 0.62 points. High-loss selection attains loss 11.913 and geometric score 0.471. The selection-stage score and the realized post-training certificate describe different quantities.

Alternative products and intervention strength. Table E.10 compares the additional products at $\gamma_{\mathrm{cf}} = 1$. The rank product yields the largest reported accuracy gain at the 1% budget among these product rules, whereas $\widetilde{\mathrm{TTI}}_\alpha$ has larger gains at 2% and 5%. Replacing loss by $p(1-p)$ gives smaller accuracy gains at every reported budget.

The retained report `reports/waterbirds_tail_transverse_influence_report.md` supplies the strength comparisons in Table E.11.

It gives numerical comparisons at $\gamma_{\mathrm{cf}} \in \{0.25, 0.5, 1.5\}$ alongside $\gamma_{\mathrm{cf}} = 1$. High loss gives larger accuracy gains at every budget for strengths 0.25 and 1.5. At $\gamma_{\mathrm{cf}} = 0.5$, $\widetilde{\mathrm{TTI}}_\alpha$ has larger gains at 1% and 2%, while high loss has the larger gain at 5%. The advantage of $\widetilde{\mathrm{TTI}}_\alpha$ at $\gamma_{\mathrm{cf}} = 1$ thus depends

Table E.10: Alternative products at $\gamma_{\mathrm{cf}} = 1$: reported mean worst-group accuracy gains (pp) and absolute ridge-certificate gains.

	Rank product		Geometric score $\times\, p(1-p)$	
Budget	Δa (pp)	ΔI	Δa (pp)	ΔI
1%	+5.45	+0.0287	+1.56	+0.0012
2%	+6.54	+0.0326	+1.71	+0.0073
5%	+6.07	+0.0516	+3.12	+0.0230

on intervention strength and budget. At $\gamma_{\mathrm{cf}} = 1.5$ and 5%, $\widetilde{\mathrm{TTI}}_\alpha$ has the larger certificate gain, although high loss has the larger accuracy gain.

Table E.11: Strength-by-budget sensitivity from the retained report named above. Entries are reported means; no seed-level uncertainty is available.

		Geometric score $\times$ loss		High loss	
γ_{cf}	Budget	Δa (pp)	ΔI	Δa (pp)	ΔI
0.25	1%	+4.05	+0.0308	+5.76	+0.0411
	2%	+7.17	+0.0366	+7.63	+0.0477
	5%	+8.88	+0.0521	+10.12	+0.0590
0.5	1%	+6.39	+0.0205	+4.36	+0.0328
	2%	+5.61	+0.0300	+4.83	+0.0446
	5%	+6.54	+0.0417	+8.57	+0.0437
1	1%	+4.52	+0.0317	+3.27	+0.0603
	2%	+6.85	+0.0452	+5.76	+0.0581
	5%	+7.94	+0.0606	+7.01	+0.0630
1.5	1%	+7.48	+0.0662	+7.63	+0.1018
	2%	+8.72	+0.0826	+9.81	+0.1006
	5%	+9.97	+0.1159	+12.31	+0.1137

Figure E.2 plots the $\gamma_{\mathrm{cf}} = 1$ accuracy and certificate gains. Connecting segments guide the eye between the three evaluated budgets.

Fitting, allocation, and aggregation. The documented run uses the `sklearn` backend with seeds 17–21. Both task head and background probe use `LogisticRegression`, `solver=lbfgs`, `max_iter=1000`, and `class_weight=balanced`; remaining settings use the library defaults, including $C = 1$, tolerance 10^{-4}, intercept fitting, and the default regularization configuration. The retained report does not pin a scikit-learn version. No Fisher-Glass penalty is used in this head refit. The base fit receives the run seed, the acquired fit receives seed $+1000$, and the shared probe receives the first run seed $+1$.

Acquired examples have no special acquisition weights. Class balancing is recomputed on the augmented set: every example of class y, original or acquired, receives class weight $N_{\mathrm{aug}}/(2N_{\mathrm{aug},y})$. The geometric matrix uses equal within-group observation weights, and the four evaluation groups have equal environment weights. The code's group-balanced control samples up to $\lceil k/4 \rceil$ examples from each resulting group in group-ID order, fills any deficit uniformly from remaining candidates, and truncates to k. Other geometric and loss selectors impose no group quotas. A random selector samples without replacement; the low-loss geometric control restricts eligibility to losses at or below the pool median.

After refitting, evaluation normalizes the new task logits using the augmented training set, while the nuisance normalization remains that of the original probe-training split. At $\alpha = 0.95$ with four equally weighted groups, the reported inverse-information tail statistic equals the maximum group inverse information. The implementation's threshold-inclusive mean and exact discrete CVaR therefore agree in value here, including tied maxima; this does not make their directional tie rules identical.

The experiment is implemented in `experiments/waterbirds_counterfactual_antidote.py`, with scoring, fitting, and aggregation helpers in `experiments/waterbirds_antidote_acquisition.py`. Its aggregator computes a mean and $1.96s/\sqrt{n}$ half-width across runs. Only aggregate reports were retained; per-seed acquisition JSON/CSV outputs are unavailable, so the tables report means without uncertainty intervals. The report describes the balanced baseline as deterministic across seeds; the seed average is not evidence from five independent benchmark splits. The numerical tables here are supported by the retained reports, rather than a fresh acquisition rerun.

Figures E.1 and E.2 expand the training and acquisition comparisons in Section 6.

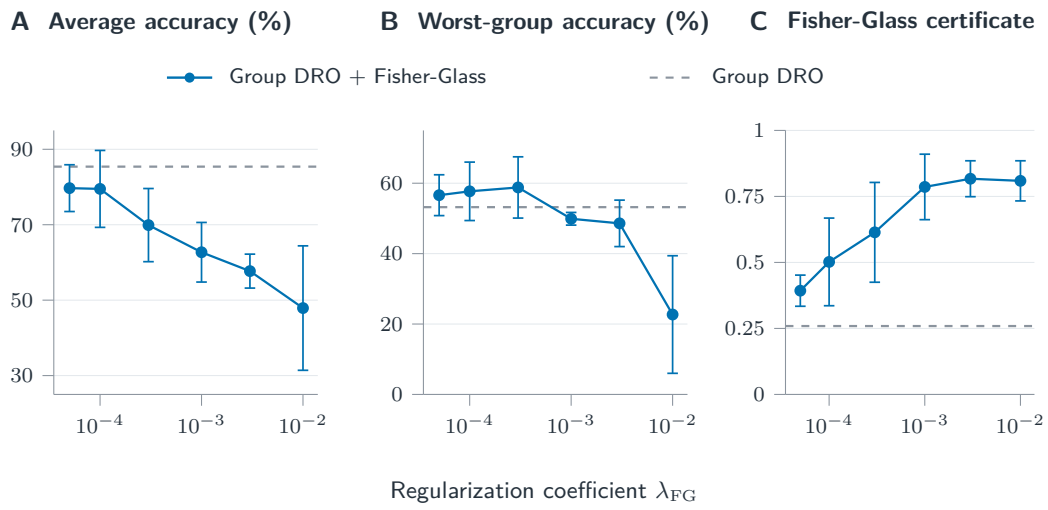


Figure E.1: **Regularization changes both information and accuracy.** End-to-end Waterbirds training with MobileNetV3-small and Group DRO plus Fisher-Glass. Points and bars show means and standard deviations across five seeds at six evaluated regularization coefficients. Dashed lines show the historical Group DRO reference from the $\lambda_{\mathrm{FG}} = 10^{-4}$ sweep cohort; it is not a common paired baseline across coefficients. Increasing the penalty raises the Fisher-Glass certificate over much of the range, while strong penalties reduce predictive accuracy. Connecting segments guide the eye.

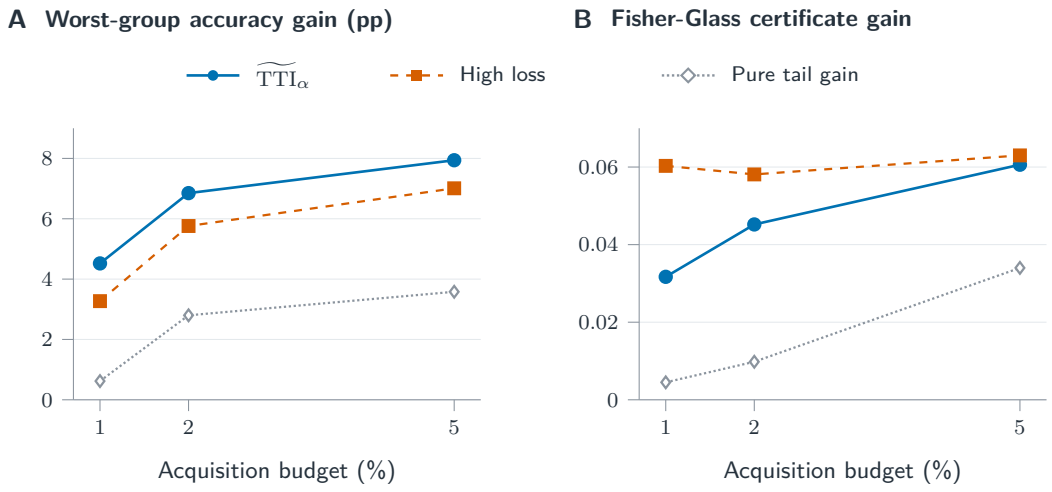


Figure E.2: **Acquisition combines a geometric score with current task loss.** Changes in held-out worst-group accuracy and Fisher-Glass certificate after feature-space counterfactual acquisition on Waterbirds, at $\gamma_{\mathrm{cf}} = 1$. The empirical $\widetilde{\mathrm{TTI}}_\alpha$ multiplies the implemented normalized residual score $\widetilde{G}_\alpha$ by current loss; the controls use high loss or the residual score alone (Appendix E.5). Points are reported means across five seeds, 17–21. Budgets are percentages of training-set size, with candidates selected from validation features. The source summary supplies no uncertainty intervals. Connecting segments guide the eye between the three evaluated budgets.

Declarations and Artifact Availability

Funding. This work received no funding.

Competing interests. The author declares no competing interests.

Code and supporting artifacts. The public repository and versioned release `camera-ready-v1` are available at

https://github.com/E-zClap/fisher-glass-learning/releases/tag/camera-ready-v1.

The release provides experiment code, dependencies, derived numerical inputs, per-seed records, checksums, and commands for reproducing the retained analyses. Its artifact manifest maps manuscript tables to the exact supporting files and identifies the corrected hidden-slice cohort. The prospective matrix and auxiliary flip-prediction reanalyses can be replayed from the included frozen numerical inputs. Original benchmark images and pretrained weights are not redistributed; dataset sources and access terms are documented separately. Historical synthetic and frozen-training summaries and the acquisition results retain their stated provenance: where original per-seed outputs are unavailable, the release includes the supporting aggregate reports and does not claim an exact replay of those missing outputs.

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